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Dual-stage deep learning framework for brain tumor classification and localization using multimodal MRI scans

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ABSTRACT

Background: Brain tumor classification and localization are essential for accurate diagnosis and effective treatment planning. With the increasing use of MRI in clinical workflows, automated tools have become crucial for assisting radiologists in providing fast and reliable analysis. Traditional approaches for tumor classification and segmentation are often time-consuming and subjective, underscoring the need for deep learning-based frameworks that can enhance diagnostic accuracy and efficiency.

Method: This study used the TCGA-GBM dataset from The Cancer Imaging Archive (TCIA) to assess a dual-stage deep learning framework combining a classification model with a RESUNET for segmentation. TCGA-GBM, part of the broader TCGA project, focuses on Glioblastoma Multiforme (GBM), a highly aggressive brain cancer. The dataset includes 3929 images, with 2556 non-tumor (class 0) and 1373 tumor (class 1) samples. The framework incorporated three convolutional neural network (CNN) architectures—MobileNet, NASNetMobile, and ResNet101—each enhanced with a Transfer Learning Layer for classification, followed by a RESUNET network for tumor localization. Transfer learning enabled the use of pre-trained weights, improving convergence speed and generalization. MobileNet offered a lightweight, efficient solution; NASNetMobile provided a strong balance between accuracy and computational cost; and ResNet101 delivered deeper feature extraction for higher precision. The RESUNET architecture, combining U-Net and residual learning, accurately segmented tumor regions, enabling effective integration of classification and localization within a unified framework.

Results: In the classification stage, the models achieved average accuracies of 0.9600 for MobileNet + Transfer Learning Layer, 0.9700 for NASNetMobile + Transfer Learning Layer, and 0.9500 for ResNet101 + Transfer Learning Layer, demonstrating their effectiveness in categorizing brain tumors. Performance was further evaluated using precision, recall, and F1 scores for both classes. MobileNet + Transfer Learning Layer achieved 0.95 precision, 0.99 recall, and 0.97 F1 for class 0, and 0.98, 0.92, and 0.95 for class 1. NASNetMobile + Transfer Learning Layer achieved 0.96, 0.99, and 0.98 for class 0, and 0.99, 0.93, and 0.96 for class 1. ResNet101 + Transfer Learning Layer achieved 0.97, 0.96, and 0.96 for class 0, and 0.93, 0.94, and 0.94 for class 1. For tumor localization, the RESUNET segmentation network accurately delineated tumor regions across all classification models.

Conclusions: The proposed dual-stage deep learning framework effectively automates both classification and localization of brain tumors from MRI scans. The results demonstrate strong performance across all architectures, with NASNetMobile + RESUNET achieving the most balanced combination of precision and efficiency. Compared to baseline CNNs and prior hybrid models, our framework achieves superior class-wise performance and more accurate tumor localization, highlighting its methodological and practical advantages. This framework shows promise for integration into real-time clinical workflows. Future research will aim to expand the approach

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to additional tumor types and MRI modalities, optimize it for real-time deployment, and validate its generalizability across diverse clinical datasets.

1. Introduction

One of the trickiest and most important issues related to neuro-oncology and medical imaging is still the detection and categorisation of brain tumors. When brain cells start to divide and expand in an irregular, dysregulated, and often uncontrollable way, departing from the body's normal cellular life cycle, brain tumors develop. These growths might be original tumors that arise in the brain tissue or they can be metastases of malignancies that started in other organs. Because of the human skull's rigid form and restricted capacity, brain tumors, regardless of their origin, present a special clinical risk. Because of compression of brain tissue, blockage of cerebrospinal fluid flow, or increased intracranial pressure, even a tiny mass in the cranial cavity may cause serious neurological impairment, in contrast to tumors in other parts of the body that can spread outside with less immediate effects. Persistent headaches, seizures, sensory deficiencies, visual problems, cognitive impairments, and in extreme situations, unconsciousness or death, are some of the possible outcomes of such pressure. Furthermore, depending on where they are located, brain tumors might impact important areas related to speech, memory, coordination, or breathing, which makes clinical care much more challenging and calls for an early, accurate diagnosis.

Brain tumors are linked to disproportionately high rates of morbidity and death, although making up just about 2% of all malignancies globally [1]. The biological aggressiveness of some tumour types, like glioblastoma multiforme (GBM), which has a median survival of less than 15 months, and the difficulty of their clinical management, which frequently calls for complex combinations of surgical, radiological, and pharmacological treatments, are both factors contributing to this increased impact. The very diverse character of brain tumors, which vary greatly in terms of development dynamics, cellular architecture, genetic composition, and treatment response, further complicates the problem [2,3]. It is very challenging to create standardised treatment regimens because of this diversity, which emphasises the urgent need for more flexible and intelligent diagnostic frameworks [4,5].

In clinical practice, precise brain tumor type and grade categorisation is crucial for directing treatment plans, predicting prognoses, and streamlining patient care. But in order to achieve this categorisation, sophisticated imaging, histological investigation, and genetic profiling must be combined—each with unique drawbacks. The need for automated, objective, and high-precision diagnostic techniques is urgent since imaging, especially MRI, is essential for non-invasive tumour visualisation, but its interpretation varies and requires skill [6].

The World Health Organisation (WHO) divides brain tumors into four classes according to molecular and histological characteristics [7, 8]. Histologically, grade I tumors mimic healthy tissue and often develop slowly. These tumors have a good prognosis and are often treatable by surgical removal. Even after initial therapy, grade II tumors often return due to their mild cellular abnormalities and infiltrative behaviour. With significant mitotic activity, cellular atypia, and a higher chance of recurrence, grade III tumors are obviously malignant. Last but not least, Grade IV tumors—like glioblastoma multiforme (GBM)—are the most aggressive; they grow quickly, invade large areas of tissue, cause necrosis, and are resistant to standard treatment. Accurately classifying these tumour categories is essential for creating therapy regimens tailored to each patient and enhancing survival rates [9].

Magnetic Resonance Imaging (MRI) serves as the cornerstone imaging modality for brain tumor diagnosis. MRI provides non-invasive, high-resolution images of soft tissues, and its various modalities—such as T1-weighted, T2-weighted, Fluid Attenuated Inversion Recovery (FLAIR), and T1-contrast enhanced (T1c)—reveal different aspects of the tumor microenvironment, including edema, necrosis, and contrast uptake. However, manual analysis of MRI scans by radiologists is subject to inter-observer variability, fatigue, and the difficulty of visualizing tumor boundaries, particularly in heterogeneous or infiltrative tumors. These limitations emphasize the need for automated, accurate, and efficient diagnostic systems that can support radiologists in clinical decision-making [10].

Despite significant advances in deep learning for brain tumor analysis, several critical gaps persist:

- Existing models rarely exploit multimodal MRI data to achieve fully integrated classification and localization within a single, unified framework.
- Current techniques often fail to robustly handle tumor class imbalance and the high variability in tumor shape and size, particularly in localization tasks.
- Many approaches are computationally intensive, requiring prolonged training times that hinder scalability and practical clinical adoption.

To overcome these limitations, the present study introduces a highly robust dual-stage framework that enhances both classification and localization accuracy while remaining computationally efficient and clinically feasible.

The proposed framework leverages the complementary strengths of deep convolutional neural networks (CNNs) for classification and semantic segmentation networks for tumor localization, enabling a fully automated end-to-end solution. The architecture comprises two integrated modules:

- Classification Module using Transfer Learning: In this phase, a pre-trained deep learning model—such as MobileNet, ResNet-101, or NASNet-Mobile—is fine-tuned using labeled MRI datasets. Transfer learning enables the model to benefit from pre-learned visual features from large-scale image datasets (like ImageNet) and adapt them to the medical domain. This component classifies MRI slices into tumor-positive and tumor-negative categories, streamlining the subsequent segmentation process.
- Segmentation Module using RESUNet: Once slices with tumors are identified, they are passed through a RESUNet model for precise pixel-wise segmentation. RESUNet combines the encoder-decoder structure of U-Net with residual learning from ResNet, allowing deeper network construction without encountering vanishing gradients. Residual connections preserve fine-grained details and improve convergence during training. By capturing both local features and global context, RESUNet is highly effective for delineating complex tumor shapes, including irregular boundaries and infiltrative edges.

This dual-pathway model ensures a comprehensive and automated pipeline that begins with tumor detection and ends with accurate boundary delineation, making it highly relevant for clinical application.

The proposed Multimodal MRI-Driven Deep Learning Framework offers an effective solution for brain tumor classification and localization by combining transfer learning-based classification with RESUNet-based segmentation. However, limitations remain, including the scarcity of annotated multimodal datasets, domain-specific generalization challenges, high computational requirements, and the lack of interpretability in deep learning models. Future improvements may involve attention mechanisms, 3D analysis, and explainable AI to boost clinical reliability and scalability.

This study presents three notable contributions to the field of brain tumor segmentation in Magnetic Resonance Imaging (MRI):

- A two-phase deep learning pipeline integrates transfer learning for binary tumor classification with RESUNet-based segmentation for detailed localization, improving both performance and computational efficiency.
- Efficiently combines spatial and semantic information from multiple MRI sequences, enhancing detection of subtle tumor variations and improving boundary delineation.
- The framework is adaptable to various backbone architectures and imaging protocols, facilitating seamless integration into diverse clinical environments.

This work introduces a novel approach characterized by the following features:

- Dual-Purpose Architecture: Classification and Segmentation: The proposed framework initially classifies MRI images as either "Tumor" or "No Tumor" using transfer learning with pre-trained convolutional neural networks, including MobileNet, NASNetMobile, and ResNet-101. If an image is classified as "Tumor", a subsequent segmentation model is employed to localize and delineate the tumor region.
- Customized Transfer Learning Layer: The segmentation model integrates a customized transfer learning component, which leverages deep encoder-decoder blocks to enhance segmentation accuracy and overall model performance.
- Validation at Two Levels: The framework incorporates distinct validation processes for both the classification and segmentation stages, ensuring a comprehensive and robust evaluation of the system's effectiveness.

This is how the rest of the paper is organised: Reviewing earlier research on brain tumor segmentation and classification, Section II (Related Work) identifies research gaps and highlights fundamental ideas. The suggested dual-stage deep learning system is presented in Section III (Methodology), which also covers its architectural layout, data preparation methods, and incorporation of multimodal MRI inputs. The experimental design, including dataset definitions, evaluation metrics, and implementation details, is described in Section IV (Experiment and Analysis). The findings achieved are then presented. The performance of the model is summed up in Section V (Result). A critical evaluation of the results, their therapeutic relevance, and a comparison with current approaches are given in Section VI (Discussion). The study's main contributions are summarised in Section VII (Conclusion), which also offers ideas for future research and model improvement.

2. Related Work

The methodology presented in Ref. [11] introduces a dual-branch MRI-based brain tumor classification model that combines ResNet50 with Multi-Head Self-Attention (MHSA) for extracting global features and VGG16 for capturing local tumor characteristics. The approach employs an attention-enhanced fusion module along with a category attention block to effectively integrate features and address class imbalance issues. Achieving a micro-average AUC of 0.989, the model surpasses existing CNN-based methods and demonstrates strong interpretability, highlighting its clinical potential. To overcome limitations of conventional models in dealing with Rician noise, image scaling, and tumor variability, the study in Ref. [12] proposes a lightweight Multi-path CNN (M-CNN). This model uses diverse convolutional filters and an optimal feature selection module. Unlike heavy pre-trained architectures such as AlexNet, ResNet, or Inception V3, the M-CNN achieves a classification accuracy of up to 96.03% with improved computational efficiency and reduced overfitting, outperforming traditional deep learning methods in brain tumor diagnosis.

In [13], the authors introduce ParMamba, a novel parallel architecture that integrates Convolutional Attention Patch Embedding (CAPE) with the ConvMamba block—combining CNN mechanisms with Mamba and a channel enhancement module. The model effectively captures both local and long-range dependencies, enabling superior tumor classification. CAPE allows efficient downsampling while maintaining rich feature representation, and the channel module enhances inter-channel interactions. ParMamba achieves up to 99.62% accuracy on benchmark datasets, significantly outperforming prior models, including Vision Transformers (ViTs), with more than 30% higher throughput. The approach in Ref. [14] presents a brain tumor classifier that fuses the Xception CNN (using 3×3 kernels) with a Parallel Deep CNN (PDCNN) employing larger kernels (5×5 and 12×12) to enhance multi-scale feature extraction. This model classifies MRI images from the Kaggle dataset into four tumor categories: meningioma, glioma, pituitary tumor, and no tumor. In Ref. [15], the authors introduce Superimposed AlexNet models (SAlexNet-1 and SAlexNet-2) for malignancy detection. These models integrate a Hybrid Attention Mechanism (HAM), dense feature extraction using 3×3 convolutions, and semi-transfer learning (STL). SAlexNet-2 achieved a classification accuracy of 99.69% in the multi-class case and 99.17% in binary classification, outperforming existing models and confirming its effectiveness in tumor analysis. The work in Ref. [16] proposes ADE_DieT, a brain tumor classification model that combines the DeiT Transformer for feature extraction with Principal Component Analysis (PCA) for dimensionality reduction and ADE for feature selection. With an accuracy of 96.09%, the model outperforms popular pre-trained models like MobileNetV3, ResNet50, and VGG16, enhancing early diagnosis and streamlining clinical workflows. The methodology described in Ref. [17] integrates a CNN model with a survival analysis component using Growth Distribution Depth (GDD). MRI scans are preprocessed using adaptive median filtering, and GDD-derived features—such as texture, shape, and gradient—are used to compute Tumor Support Factor (TSF) and tumor weight (TW). The model achieves up to 97% accuracy while also estimating patient survival rates. The study in Ref. [18] presents a two-stage approach utilizing the Br35H MRI dataset. Initially, image quality is enhanced using GFPGAN and Real-ESRGAN. Then, nine deep learning models are trained, and ensemble learning is applied in the second stage. The approach achieved 100% accuracy with the GFPGAN-enhanced dataset and top-performing classifiers, demonstrating strong potential for reliable tumor classification.

In [19], the authors propose a deep learning strategy for early-stage tumor detection, utilizing a CNN enhanced with transfer learning to process diverse medical imaging data. With optimized preprocessing and hyperparameters, the model achieves robust results across key performance metrics, supporting early tumor growth prediction and improved treatment planning. The research in Ref. [20] explores brain cancer diagnosis using deep learning on MRI datasets. While pre-trained models initially achieved 100% across all evaluation metrics on a primary BT-MRI dataset, performance declined when tested on synthetic datasets. Integrating data augmentation techniques like Gaussian noise and blurring during training significantly improved generalization and diagnostic robustness. The study in Ref. [21] combines machine learning with transfer learning to classify healthy and affected individuals using both numerical and image data. Using 3762 MRI images, the modified GoogLeNet model reached 99.3% classification accuracy, while LightGBM yielded 95.7% on numerical features, illustrating the strength of multimodal learning approaches. A novel multi-task network is introduced in Ref. [22], combining a modified VGG16 with U-Net to perform simultaneous segmentation, classification, and localization of brain tumors. Trained on two datasets, the model achieved 97% classification accuracy and a Dice score of 0.86 for segmentation, offering high efficiency for use in low-resource healthcare environments.

The work in Ref. [23] proposes a lightweight EfficientNet (ECNN) model for brain tumor detection using transfer learning. By classifying

tumors into meningioma, glioma, and pituitary types across three datasets, the ECNN method outperforms others, achieving an average accuracy of 98.48%, with high recall, precision, and sensitivity. In Ref. [24], a Support Vector Machine (SVM) approach is presented for classifying brain tumors based on preprocessed MRI images. Through median filtering, Otsu's thresholding, and GLCM-based feature extraction, the model achieves 99.9% accuracy on a small dataset, outperforming prior methods. The authors of [25] introduce a Computer-Aided Diagnosis (CAD) system integrating CNN, VGG-19, and LSTM networks for segmenting and classifying tumors in MRI images. The system distinguishes between slow-growing and aggressive tumors, achieving 97.98% accuracy across five datasets and supporting early diagnosis with reduced treatment costs.

The research in Ref. [26] evaluates deep transfer learning models—InceptionV3, VGG19, DenseNet121, and MobileNet—for classifying brain tumors into four categories. After data augmentation to balance classes, MobileNet achieved the highest accuracy of 99.60%. In Ref. [27], the proposed Support Value-based Deep Neural Network (SDNN) detects tumors in brain images for IoMT (Internet of Medical Things) environments. Key features like entropy and texture are extracted post skull-stripping preprocessing. The SDNN outperforms conventional methods in accuracy and efficiency. EMU-Net, proposed in Ref. [28], is a modified U-Net architecture applied to the BraTS 2020 dataset for brain tumor classification and segmentation. It incorporates GLCM for feature extraction, CNN for classification, and chi-square for feature selection, achieving 99.65% accuracy and high Dice scores for different tumor regions. The RBEBT model in Ref. [29] uses a fine-tuned ResNet18 for feature extraction, a randomized neural network (RNN) for classification, and a bat algorithm for optimization. Achieving 99% accuracy and 100% sensitivity via five-fold cross-validation, it outperforms conventional CNN-based approaches. The study in Ref. [30] introduces a DenseNet121-based model trained with CycleGAN-generated synthetic images to improve classification accuracy for glioma, meningioma, and pituitary tumors. This innovative use of synthetic data improves generalization and outperforms models trained on real data alone.

Study [31] proposed a Least Square–Support Vector Machine (LS-SVM) based brain tumor classification system that integrates multi-model texture features to improve classification performance. Their study, emphasized the importance of combining multiple texture descriptors derived from MRI images to capture both spatial and intensity variations within tumor regions. By optimizing the LS-SVM classifier with these diverse features, the model demonstrated superior classification accuracy and robustness compared to traditional single-feature approaches. This highlights the effectiveness of texture-based feature fusion in improving model generalization across varied MRI datasets. In another study [32], introduced MRI-based ensemble frameworks designed to enhance the prediction of human brain tumors through multi-model learning integration. Their research, developed several ensemble configurations combining multiple base learners, including convolutional neural networks (CNNs) and classical machine learning models. The ensemble methodology achieved improved accuracy and stability by leveraging the strengths of different models, effectively reducing bias and variance in tumor classification tasks. Complementing these approaches [33], developed a Weighted Average Ensemble Deep Learning model aimed at stratifying brain tumors in MRI images. This study introduced a framework that assigns adaptive weights to multiple deep neural networks based on their predictive reliability. The weighted ensemble strategy ensures that more accurate models contribute more significantly to the final decision, thereby enhancing classification precision. Their experimental evaluation showed that the weighted ensemble outperformed conventional single-model deep learning architectures in terms of accuracy, sensitivity, and specificity.

3. Methodology

3.1. MobileNet architecture as a base model

A lightweight and computationally effective deep neural network architecture, MobileNetV1 was created especially for embedded and mobile vision applications. By drastically lowering computational complexity while preserving excellent accuracy, it tackles the difficulty of implementing deep learning models in situations with limited resources. Depthwise Separable Convolutions, which break down a typical convolution into two operations—depthwise convolution (spatial filtering per input channel) and pointwise convolution (combining channel-wise outputs via 1×1 convolutions)—are the main invention of MobileNetV1. Significant efficiency benefits are achieved without a significant performance reduction because to this factorisation, which lowers the number of trainable parameters and floating point operations per second (FLOPs) [34].

With their many parameters and thick convolutional layers, traditional deep neural networks like VGG-16 and ResNet-50 demand a lot of processing power, which makes them unsuitable for mobile and edge devices. By using a parameter-efficient design that lowers memory footprint, power consumption, and inference latency, MobileNetV1 addresses this problem and is thus well suited for deployment on low-power hardware, including smartphones, Internet of Things devices, and edge AI systems. Furthermore, MobileNetV1 includes two crucial scalability hyperparameters: the resolution multiplier (ρ), which modifies input image resolution to maximise computational efficiency, and the width multiplier (α), which regulates the number of channels per layer to trade off accuracy for model size [35].

With far fewer parameters than traditional CNNs, MobileNetV1 has great generalisation capabilities and performs competitively on large-scale datasets such as ImageNet thanks to its architectural optimisations [35]. For real-time picture classification, object identification, face recognition, and other computer vision tasks in mobile and edge computing settings, the model's processing efficiency makes it very useful. For deep learning applications in resource-constrained contexts, MobileNetV1 continues to be a popular choice since it balances accuracy, latency, and energy economy.

3.1.1. Depthwise separable convolution

Context and feature extraction from the incoming picture are the encoder's responsibilities. This is made up of many convolutional blocks, with two 3x3 convolutions in each, a rectified linear unit (ReLU), and a 2x2 max pooling operation with stride 2 for downsampling coming after. More complicated characteristics may be captured at lower spatial resolution since each block doubles the number of feature channels [34]. Mathematically, each layer can be represented as:

I. **Depthwise Convolution** – Applies a single filter per input channel, processing each channel independently.
II. **Pointwise Convolution (1×1 Conv)** – Applies 1×1 convolutions to merge the outputs from depthwise convolution into the desired number of output channels.

By decoupling spatial filtering and channel mixing, Depthwise Separable Convolutions achieve significant reductions in computational cost and model size. A standard convolution operation is performed in a single step, where a convolutional filter is applied to all input channels simultaneously to produce an output feature map [36]. Mathematically, the computational cost of a standard convolution is given by:

$$FLOPs = H' \times W' \times C_{in} \times C_{out} \times K^2$$

Where H' and W' represent the height and width of the output feature map, C_{in} is the number of input channels, C_{out} is the number of output channels, and K^2 is the kernel size.

Each convolutional filter of size K^2 slides over the entire input feature map and aggregates information across all input channels. This results in a high computational cost, especially when the number of input and output channels increases. For instance, if an input layer has 32 channels ($C_{in} = 32$) and the output layer has 64 channels ($C_{out} = 64$), then each output feature map is computed by combining contributions from all 32 input channels [36]. Given that a convolution operation is performed for every location in the feature map, the number of operations grows significantly, making standard convolution computationally expensive.

Depthwise Separable Convolutions figure [7] split the process into two steps, reducing redundant computations:

I. Depthwise convolution is a fundamental component of depthwise separable convolutions, a technique used in efficient neural network architectures like *MobileNet*. Unlike standard convolution, which applies multiple filters to all input channels simultaneously, depthwise convolution processes each input channel independently using a single spatial filter [36]. This approach dramatically reduces the number of computations while preserving spatial information. In a standard convolution operation, each convolutional kernel (of size K × K) is applied across all input channels at once, leading to a large number of multiplications. However, in depthwise convolution, the operation is broken down into two steps:
   a. Channel-wise Processing
      - Each input channel has its own separate convolutional filter of size K × K.
      - Instead of aggregating information across multiple channels, each filter operates only on its corresponding input channel.
      - This allows the convolution to maintain spatial relationships within each channel while avoiding excessive computation.
   b. Reduction in Computational Cost

Since a separate $K \times K$ filter is applied to each channel instead of the entire input volume, the number of required multiplications is significantly lower.

Mathematically, the FLOP (floating point operation) count for depthwise convolution is given by:

$$\textit{depthwise Conv FLOPs} = H' \times W' \times C_{in} \times K^2$$

where:

- H′ and W′ are the height and width of the output feature map.
- C_{in} is the number of input channels (which remains unchanged in depthwise convolution).
- K × K is the kernel size.

Depthwise convolution has intrinsic limits that impact its representational capabilities, even if it is computationally efficient. Because each filter works independently on a single channel, it does not allow interaction between several feature channels, which is one of its main disadvantages. Complex inter-channel relationships are essential for learning high-level feature representations, but this limitation prevents the model from capturing them. A pointwise convolution (1 × 1 convolution) is added after the depthwise operation in depthwise separable convolution to overcome this restriction [36]. By permitting cross-channel interactions, the pointwise convolution successfully restores feature expressiveness by projecting the depthwise-convolved feature maps into a new space. This combination preserves the network's capacity to learn rich and significant feature representations while maintaining computational efficiency [34].

II. **Pointwise Convolution:** After applying depthwise convolution, each channel in the feature map is processed independently. While this reduces computational complexity, it also introduces a limitation: no interaction occurs between different channels. This is where pointwise convolution (1 × 1 convolution) comes in—it is responsible for combining information across channels while maintaining the spatial structure of the feature map. A *pointwise convolution* is simply a 1 × 1 convolutional filter applied to each pixel in the feature map across all channels. Unlike standard convolutions that use larger kernels (e.g., 3 × 3 or 5 × 5), the 1 × 1 kernel does not perform spatial filtering—it only mixes information between channels [36].
   a. Each pixel in the feature map has multiple channels (depth).
   b. A 1 × 1 filter is applied across all input channels for each pixel.
   c. Each 1 × 1 filter outputs a single value per pixel by linearly combining the input channels.
   d. Multiple 1 × 1 filters are used to create multiple output channels.

Mathematically, the FLOP (floating point operations) count for pointwise convolution is:

$$\textit{Pointwise Conv FLOPs} = H' \times W' \times C_{in} \times C_{out}$$

Where:

- H' and W' are the height and width of the output feature map.
- C_{in} is the number of input channels (from depthwise convolution).
- C_{out} is the number of output channels (i.e., the number of 1 × 1 filters applied).

Pointwise convolution plays a crucial role in depthwise separable convolution by efficiently mixing information across channels. Since depthwise convolution has already extracted spatial features, the 1 × 1 convolution focuses solely on integrating inter-channel dependencies, ensuring effective feature learning. This approach significantly lowers computational cost by eliminating redundant spatial computations present in standard convolutions, thereby reducing the number of floating-point operations (FLOPs) [36]. Additionally, pointwise convolution enhances the model's expressiveness by assigning unique weights to different channels, allowing the network to dynamically emphasize important features and improve its ability to learn complex patterns. Without pointwise convolution, depthwise convolution alone would lack the capability to capture meaningful multi-channel feature representations, making it an essential component of depthwise separable convolution [34] (see Fig. 1).

The MobileNetV1 architecture is shown in Fig. 2.

3.2. NASNet-mobile architecture as a base model

NASNet (Neural Architecture Search Network) is a deep learning model developed by Google Brain. The goal of NASNet is to automatically design neural networks using Neural Architecture Search (NAS). Neural Architecture Search (NAS) is an advanced technique in machine learning that automates the design of neural network architectures, eliminating the need for manual configuration [37]. Traditionally, deep learning models require human experts to define key structural components such as the number of layers, types of operations, and the way different layers connect. However, NAS algorithms replace this manual process by systematically exploring a predefined search space of potential architectures to identify the most optimal design for a given task. This search space includes various layer types, convolutional operations, skip connections, and activation functions, allowing NAS to evaluate countless possible combinations [38].

One way to understand NAS is by comparing it to hyperparameter optimization. While traditional hyperparameter tuning focuses on adjusting values like learning rate, batch size, or dropout rate within a fixed architecture, NAS takes optimization a step further by searching for the best network structure itself. This distinction is significant because hyperparameter tuning refines an existing model, whereas NAS

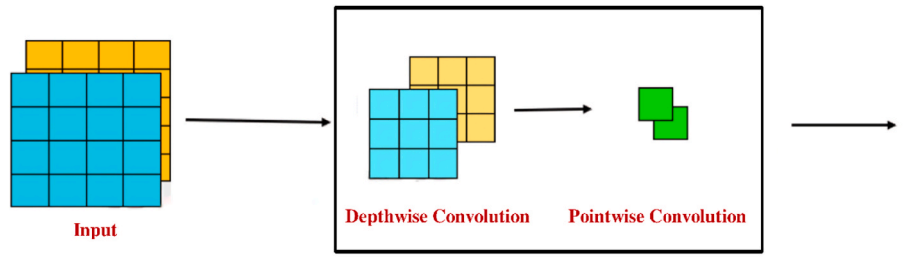


Fig. 1. Depthwise separable convolution.

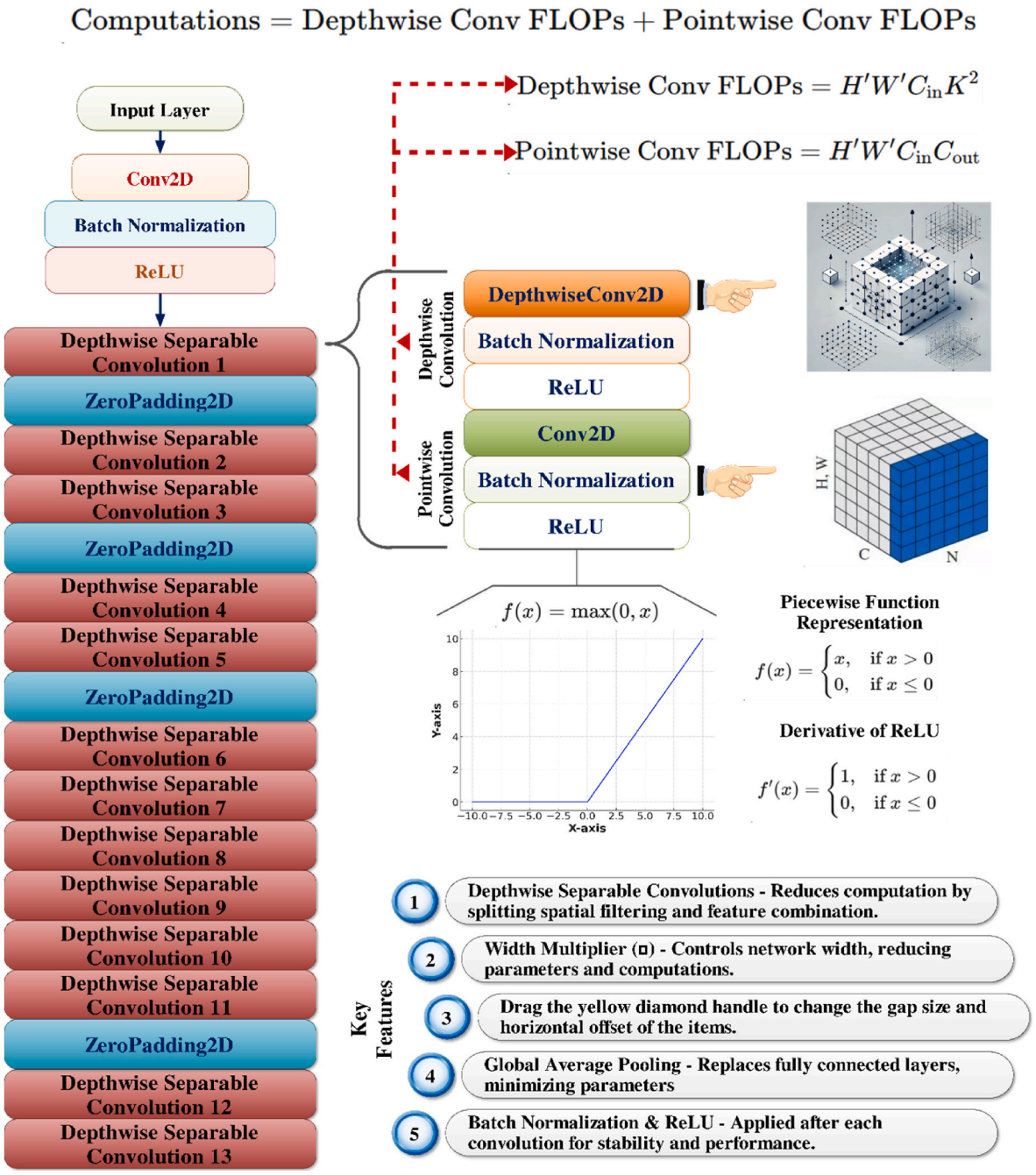


Fig. 2. MobileNetV1 architecture.

has the potential to discover entirely new architectures that may outperform manually designed models [37]. By leveraging

reinforcement learning, evolutionary algorithms, or gradient-based search methods, NAS systematically evaluates different architectural configurations, selecting those that yield the best performance.

The NASNetMobile architecture [Fig. 4] consists of four main components [Fig. 3]:

The **Stem Network** is the first stage of *NASNet* and is responsible for processing the input image to extract low-level features such as edges, textures, and simple patterns. It consists of several convolutional layers that reduce the input image size while preserving important features.

How It Works:

- The input image undergoes a series of standard convolutions and ReLU activations to extract initial patterns.
- These convolutions convert raw pixel data into feature maps, which are later processed by *Normal* and *Reduction Cells*.
- The output of the Stem Network is a set of feature maps that contain important visual information, which will be further refined in the next layers.

A convolution operation in the Stem Network can be written as:

$$Y(i,j,k) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} \sum_{c=0}^{C-1} [X(i+m, j+n, c) \cdot W(m,n,c,k)] + B(k)$$

where:

- X(i,j,c) → Input image or feature map
- W(m,n,c,k) → Convolutional kernel (filter)
- B(k) → Bias term
- Y(i,j,k) → Output feature map

The **Normal Cell** is responsible for learning complex patterns while maintaining the spatial dimensions of the feature maps. This means that the height and width of the input feature maps remain unchanged, while new features are extracted.

How it works:

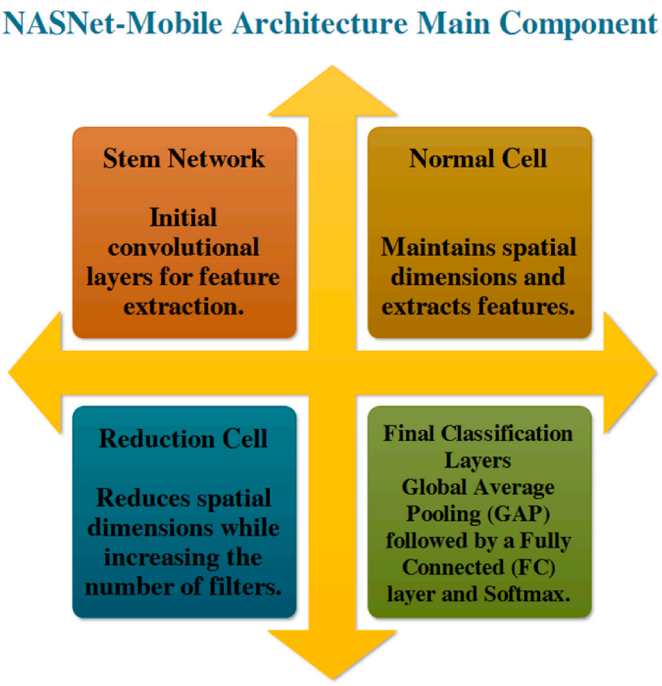


Fig. 3. NASNet-mobile architecture main component.

- The Normal Cell consists of a combination of operations such as separable convolutions, dilated convolutions, max pooling, and addition operations.
- These operations enhance feature extraction without reducing the size of the feature maps.
- Since the spatial dimensions remain unchanged, Normal Cells focus on improving feature richness rather than downsampling.

Operations in a Normal Cell:

1) Separable Convolution: Reduces computation while preserving accuracy.

$$Y_d(i,j,k) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} [X(i+m, j+n, k) \cdot W_d(m,n,k)]$$

2) Skip Connections: Preserve important gradients.

$$Y = X + f(x)$$

3) Concatenation of Outputs: Different transformations are combined.

$$Y = Concat(f_1(X), f_2(X))$$

The **Reduction Cell** is designed to reduce the spatial dimensions (height & width) of the feature maps while increasing the number of filters. This is crucial because as the network goes deeper, it needs to reduce computational complexity while maintaining meaningful features.

How It Works:

- Instead of simple downsampling like max pooling, the Reduction Cell applies strided convolutions or dilated convolutions to reduce spatial resolution while increasing depth.
- This ensures that as the model becomes deeper, it captures more abstract patterns.
- The number of filters doubles to ensure more detailed representations are learned at each stage.

Representation of a Strided Convolution:

$$Y(i,j,k) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} [X(2i+m, 2j+n, c) \cdot W(m,n,c,k)] + B(k)$$

Where $2i$ and $2j$ indicate a stride of 2, which reduces the feature map's size by half.

After passing through multiple Normal and Reduction Cells, the extracted features are fed into the final classification layers to make a prediction.

Key Components:

- Global Average Pooling (GAP): Instead of using fully connected layers, NASNet uses Global Average Pooling to reduce each feature map to a single value. This reduces parameters and improves generalization.

$$Y_k = \frac{1}{H \times W} \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} X(i,j,k)$$

- Fully Connected (FC) Layer: The pooled features are fed into a fully connected layer, which maps the features to the number of output classes. If we have N classes, the final output is a vector of length N.
- Softmax Activation: The final output is passed through a softmax function, which converts raw scores into probabilities.

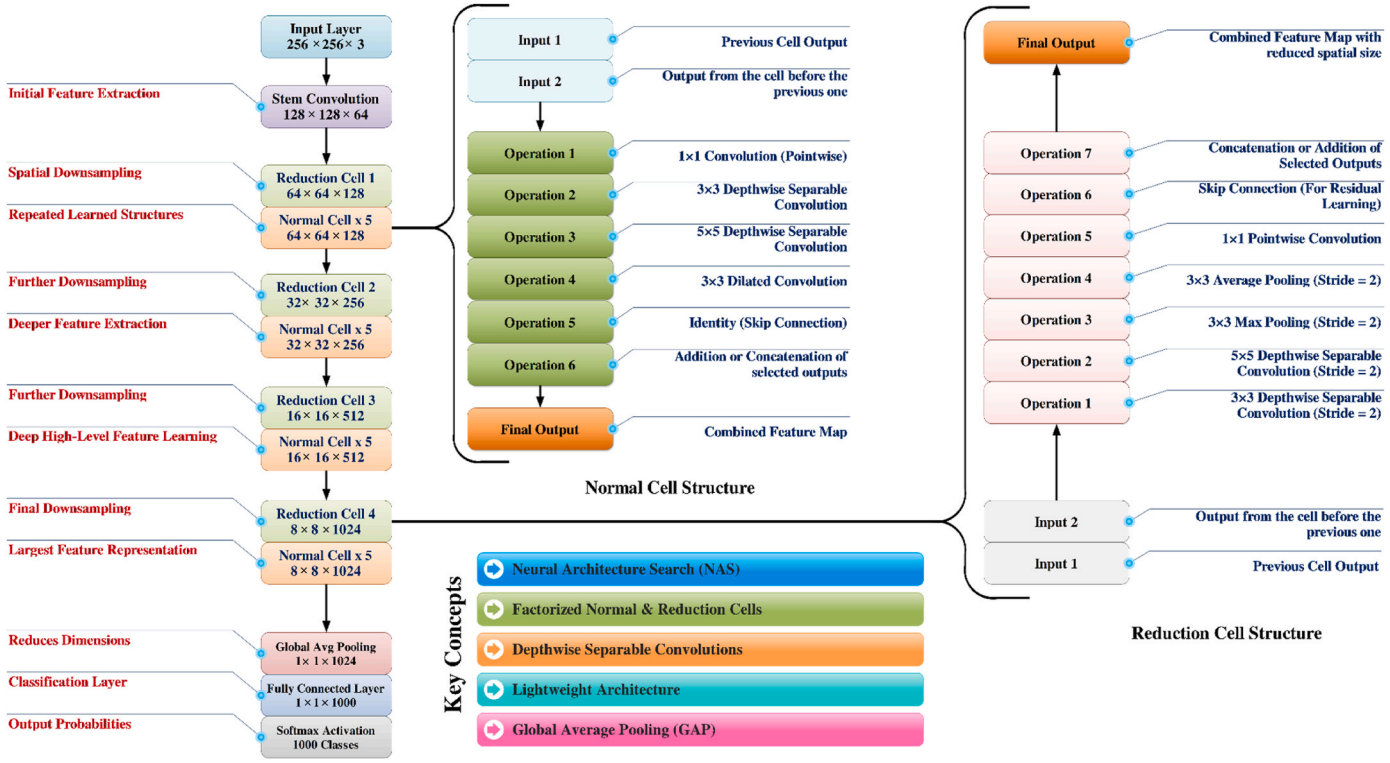


Fig. 4. NASNet-mobile architecture.

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^{N} e^{z_j}}$$

where z_i is the logit value for class i.

3.3. RESNET-101 architecture as a base model

ResNet-101 (Residual Network with 101 layers) is a deep convolutional neural network designed to address the challenges associated with training very deep networks [39]. Before ResNet, increasing the depth of neural networks often led to the *vanishing gradient problem*, where gradients became too small to effectively update earlier layers during backpropagation. This resulted in difficulty training deeper networks, limiting their performance. ResNet-101 introduced the concept of residual learning using skip connections, which allows gradients to flow more easily through the network, improving both training efficiency and accuracy [40]. Deep networks are essential for extracting complex hierarchical features from images, but simply stacking more layers does not always lead to better performance. Traditional deep networks suffer from two major issues:

- **Vanishing Gradient Problem:** As gradients propagate back through many layers, they become too small to significantly update the earlier layers. This makes deeper networks ineffective.
- **Degradation Problem:** When adding more layers, accuracy should ideally improve. However, in practice, very deep networks often perform worse than their shallower counterparts due to ineffective gradient flow.

To solve these issues, ResNet introduced skip connections, also known as identity shortcuts, which allow the network to directly pass information from earlier layers to later layers. Instead of learning a direct mapping $H(x)$, ResNet learns a residual function:

$$H(x) = F(x) + x$$

Here, $F(x)$ represents the transformations (convolutions, batch normalization, and activation functions) applied to the input x. By adding the original input x to the output of these transformations, the network ensures that even if the transformation $F(x)$ becomes small, the identity mapping x remains, helping gradients to flow smoothly through the network.

ResNet-101 [Fig. 5] consists of 101 layers organized into five stages, each containing multiple bottleneck blocks. The first stage consists of a standard convolutional layer followed by a max-pooling layer, while the remaining four stages consist of residual blocks with skip connections.

3.3.1. Bottleneck blocks: the core of ResNet-101

The key innovation in ResNet-101 is the bottleneck block, which consists of three convolutional layers instead of the two used in standard residual blocks. This design reduces computational complexity while maintaining performance.

Each bottleneck block contains the following components:

- **1 × 1 Convolution (Dimensionality Reduction):**\

This reduces the number of channels before performing the main 3 × 3 convolution, decreasing the computational load. Given an input feature map X of size $H \times W \times C_{\text{in}}$, a 1 × 1 convolutional filter with C_{out} output channels applies the following transformation:

$$Y = W \times X + b$$

where:

- $X \in R^{H \times W \times C_{in}}$ – input tensor
- $W \in R^{1 \times 1 \times C_{in} \times C_{out}}$ – weight of the 1 × 1 convolution filter
- $b \in R^{C_{out}}$ – bias term
- $Y \in R^{H \times W \times C_{out}}$ - output tensor

Each output channel is computed as:

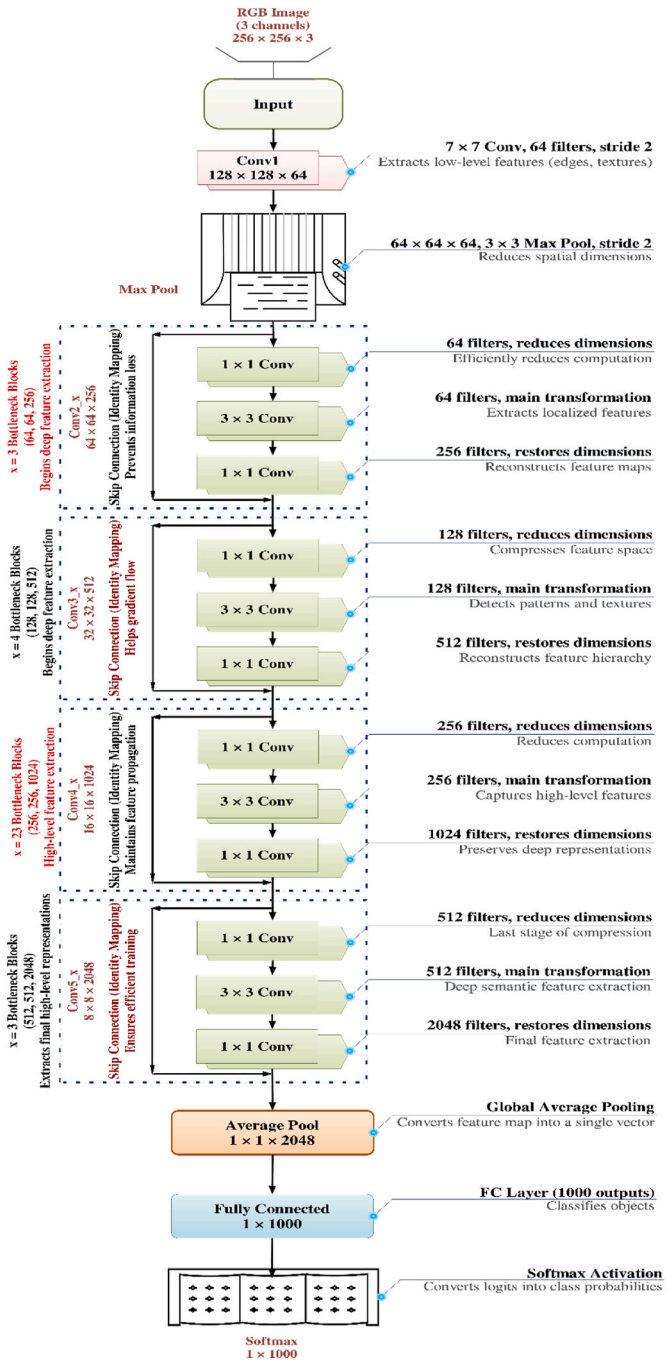


Fig. 5. RESNET-101 architecture.

$$Y_{h,w,k} = \sum_{i=1}^{C_{in}} \left(W_{i,k} \cdot X_{h,w,i} + b_k \right)$$

where:

- h, w - spatial position in the feature map
- k - output channel index
- i- input channel index

- **3 × 3 Convolution (Feature Extraction):**

This is the core transformation step, responsible for spatial feature extraction. Given X of size H × W × C_{in}, the 3 × 3 × C_{in} × C_{out} to produce Y:

$$Y_{h,w,k} = \sum_{i=1}^{C_{in}} \sum_{m=0}^{2} \sum_{n=0}^{2} \left(W_{m,n,i,k} \cdot X_{h+m,w+n,i} + b_k \right)$$

where:

- h, w - spatial position in output feature map
- i - input channel index
- k - output channel index
- $W_{m,n,i,k}$ - weight of the kernel at position (m,n)
- $X_{h+m,w+n,i}$ - input value at corresponding location
- b_k - bias term for the kth output channel

- **1 × 1 Convolution (Dimensionality Restoration):**

This restores the number of channels to the original count before adding the residual connection. Let input X be a size H × W × $C_{reduced}$. A 1 × 1 filter restores it to H × W × $C_{original}$:

$$Y = W \times X + b$$

where:

- $X \in R^{H \times W \times C_{in}}$ – input tensor
- $W \in R^{1 \times 1 \times C_{in} \times C_{out}}$ – weight of the 1 × 1 convolution filter
- $b \in R^{C_{out}}$ – bias term
- $Y \in R^{H \times W \times C_{out}}$ - output tensor

Each output is computed as:

$$Y_{h,w,k} = \sum_{i=1}^{C_{reduced}} \left(W_{i,k} \cdot X_{h,w,i} + b_k \right)$$

where:

- h, w - spatial position
- k - output channel index
- i- input channel index
- $W_{i,k}$ – weight assigned to the ith input channel for the kth output channel

Each bottleneck block also includes a skip connection, which bypasses the three convolutional layers and adds the original input directly to the output. This helps maintain gradient flow, enabling deep networks to train effectively. It is also called a shortcut connection that is a technique where the input x of a layer is directly added to its output, bypassing the main transformation function. Mathematically, this can be written as:

$$Y = F(X) + X$$

where:

- X - is the input to the residual block.
- $F(X)$ - is the transformation applied by convolutional layers.
- Y - is the final output after the skip connection.

3.4. Added deep learning model with transfer learning

This study presents the architectural modifications applied to a pre-trained deep neural network utilizing the principles of transfer learning. The primary objective is to enhance the network's classification capabilities by appending a custom classification head while preserving the pre-trained feature extraction layers [Fig. 6]. First, the base model is frozen, meaning its pre-trained weights remain fixed and are not updated during training. This ensures that the previously learned features are retained while training only the newly added classification layers. The output of the base model, denoted X_{base}, serves as the input for these additional layers. Mathematically,

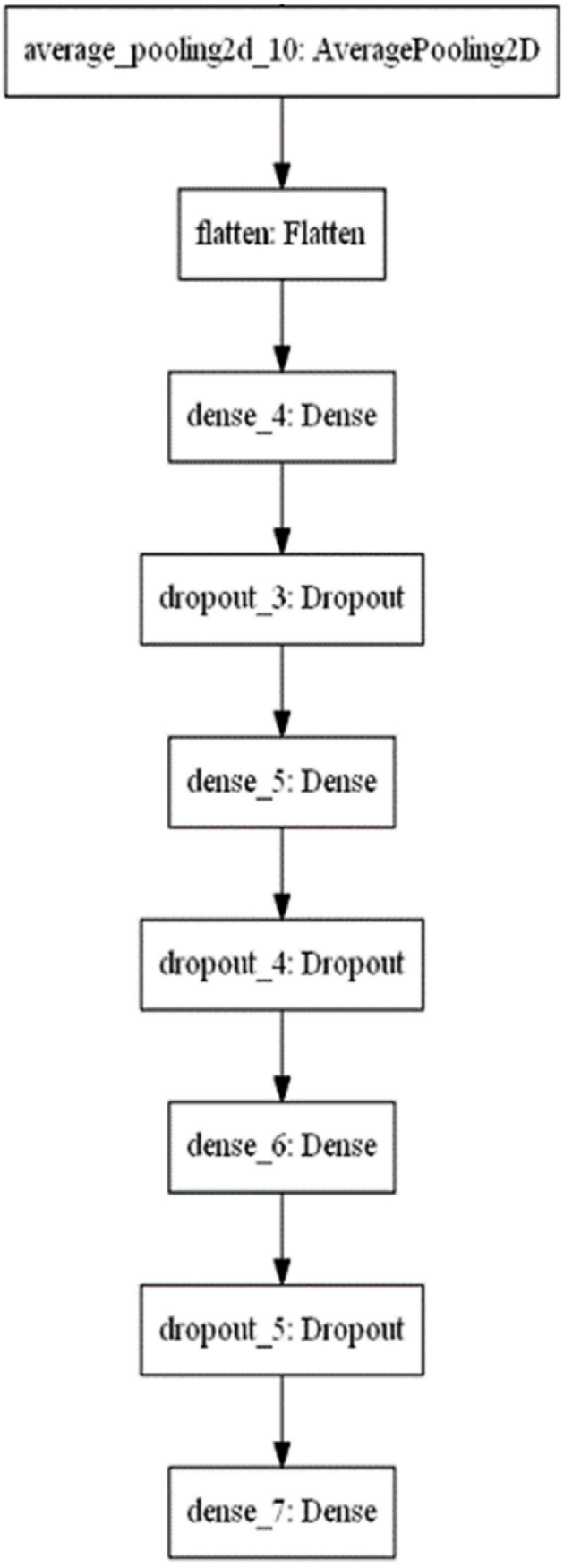


Fig. 6. Transfer learning layer.

$X_{base} \in R^{h \times w \times c}$

where h represents the height, www the width, and c the number of channels (depth) of the feature maps.

To enhance feature generalization and reduce computational complexity, an AveragePooling2D layer is applied with a kernel size of 4 × 4. This operation computes the mean of local regions within the feature maps, as given by:

$$X_{pool}(i,j) = \frac{1}{k^2} \sum_{m=0}^{k-1} \sum_{n=0}^{k-1} X_{base}(i+m, j+n)$$

where i and j represent the spatial locations in the output feature map, and the pooling layer reduces the dimensions to $R^{\frac{h}{k} \times \frac{w}{k} \times c}$. Next, a Flatten layer transforms this three-dimensional tensor into a one-dimensional vector:

This flattened representation is then fed into three fully connected (Dense) layers, each comprising 256 neurons and activated using the ReLU function to introduce non-linearity. The first dense layer applies a transformation:

$$X_1 = ReLU(W_1 \cdot X_{flatten} + b_1)$$

Where $W_1 \in R^{256 \times d}$ and $b_1 \in R^{256}$ represent the weight matrix and bias vector, respectively. The output size after this transformation is R^{256}. To mitigate overfitting, a Dropout layer with a rate of 25% is introduced, randomly deactivating neurons during training. This is expressed mathematically as:

$$X_{drop} = X_1 \odot R \; where \; R \in \{0,1\}^{256}, E[R] = 0.75$$

where $\odot$ represents element-wise multiplication, and R is a random vector with 25% of values set to 0. Two additional dense layers follow the same transformation:

$$X_2 = ReLU(W_2 \cdot X_{drop} + b_2)$$

$$X_3 = ReLU(W_3 \cdot X_2 + b_3)$$

Each layer maintains an output of dimension R^{256} , followed by another dropout operation. Finally, the output layer consists of two neurons with a Softmax activation function, computing the probability of each class:

$$X_{out} = Softmax(W_4 \cdot X_{drop} + b_4)$$

Where $W_4 \in R^{256 \times 2}$ and $b_4 \in R^2$. The Softmax function normalizes the outputs so that their sum equals 1:

$$Softmax(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{2} e^{z_j}}, for \; i = 1, 2$$

Thus, the model effectively maps an input X_{base} of dimensions $R^{h \times w \times c}$ to an output $Y \in R^2$ making it suitable for binary classification. By leveraging a pre-trained base model and integrating a new classification head, this approach optimally balances computational efficiency and classification accuracy, demonstrating a robust application of transfer learning.

Step by Step Approach for Transfer Learning-Based CNN with Custom Classifier

Input:

$I \in R^{256 \times 256 \times 3}$: Input image tensor

$M \in \{NASNetMobile, MobileNet, ResNet101\}$: Base model Pre-trained ImageNet weights

Output:

Final model F_{final} : $R^{256 \times 256 \times 3} \rightarrow R^2$ for binary classification

Procedure:

Step 1: Load Pre-Trained Base Model

Select a CNN base $B(I)$ such that:

(*continued on next page*)

(*continued*)

Step by Step Approach for Transfer Learning-Based CNN with Custom Classifier
$X_{base} = B(I)$ $X_{base} \in R^{h \times w \times c}$ where h, w, are spatial dimensions and c is the number of feature channels. The base excludes the top classifier. **Step 2: Freeze Base Layers** For each layer $L_i \in B$, set: $trainable(L_i) = False$ This prevents updates to pre-trained weights during backpropagation. **Step 3: Apply Average Pooling** $X_1(i,j,k) = \frac{1}{k^2}\sum_{m=0}^{k-1}\sum_{n=0}^{k-1} X_{base}(i+m, j+n, k)$ Where k = 4 is the pooling window size. Output becomes: $X_1 \in R^{h' \times w' \times c}, h' = \lceil h/4 \rceil, w' = \lceil w/4 \rceil$ **Step 4: Flatten Feature Maps** $X_{flat} = reshape(X_1) \in R^d, d = h' \cdot w' \cdot c$ **Step 5: Add Fully Connected Layers** Each dense layer performs: $X^{(l)} = ReLU\left(W^{(l)}X^{(l-1)} + b^{(l)}\right), l = 1,2,3$ $W^{(l)} \in R^{256 \times d}, b^{(l)} \in R^{256}, X^{(0)} \in X_{flat}$ **Step 6: Apply Dropout (Regularization)** For each dense output $X^{(l)}$, apply dropout: $X^{(l)}_{drop} = X^{(l)} \odot R^{(l)}, R^{(l)} \in \{0,1\}^{256}, E\left[R^{(l)}\right] = 0.75$ where $\odot$ is element-wise multiplication and 25% of neurons are randomly zeroed. **Step 7: Final Output Layer** $Z = W^{(4)}X^{(3)}_{drop} + b^{(4)}, W^{(4)} \in R^{2 \times 256}, b^{(4)} \in R^2$ $\hat{y} = Softmax(Z), \hat{y}_i = \frac{e^{Z_i}}{\sum_{j=1}^{2} e^{Z_j}}, i = 1,2$ **Step 8: Define Full Model** $F_{final} = Model(I \rightarrow \hat{y})$

3.5. RESUNET architecture for localization

RESUNet is a hybrid deep convolutional neural network architecture [Fig. 7] that integrates the structural advantages of U-Net with the representational power of residual learning as introduced in ResNet. Originally developed for biomedical image segmentation, the U-Net architecture employs a symmetric encoder-decoder topology with skip connections that enable the model to leverage both high-level contextual information and low-level spatial details. However, as network depth increases, conventional convolutional models become susceptible to optimization difficulties, such as vanishing gradients and performance degradation [41].

To address these limitations, RESUNet introduces residual blocks within both the encoder and decoder paths. Each residual unit consists of two or more convolutional layers, interleaved with batch normalization and ReLU activations, and incorporates identity mappings via additive skip connections. These residual pathways enable more efficient gradient flow during backpropagation and support the learning of residual functions rather than direct mappings, which has been shown to accelerate convergence and improve generalization in deeper architectures.

The encoder path in RESUNet progressively reduces the spatial resolution of feature maps while increasing their depth, capturing hierarchical semantic features [41]. This is achieved through sequential residual blocks followed by downsampling operations such as max pooling. At the bottleneck—the deepest layer in the network—high-level abstract representations are learned. The decoder path then performs spatial upsampling, typically through transposed convolutions, to restore the original input resolution. Crucially, feature maps from the encoder are concatenated with the decoder via long-range skip connections, preserving spatial fidelity and enabling precise localization. Residual blocks are also employed in the decoder to refine these fused representations [42].

The final layer applies a 1 × 1 convolution to project the feature maps into the desired number of output classes, followed by a sigmoid or softmax activation function, depending on whether the task is binary or multi-class segmentation. Overall, RESUNet achieves a balance between depth and spatial accuracy, making it particularly well-suited for tasks that require both global contextual understanding and detailed localization, such as medical image analysis, remote sensing, and automated defect detection [42].

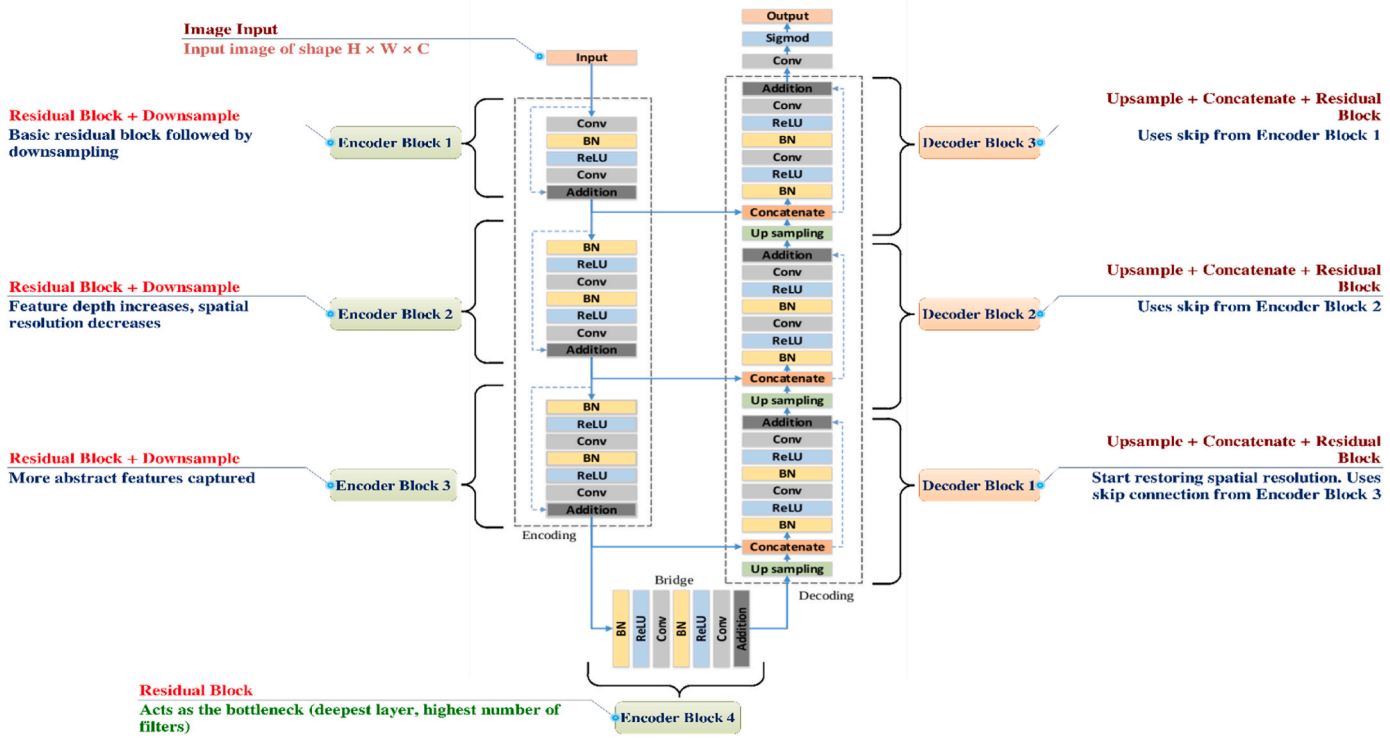


Fig. 7. RESUNET architecture walkthrough.

3.6. Overall framework for classification and localization

For precise brain tumor identification and localization from MRI images, the system shown combines deep learning-based classification and segmentation. InceptionV3, ResNet101, DenseNet201, and other pre-trained convolutional neural networks (CNNs) are used as backbone models in this two-phase design, which starts with a categorisation model. To differentiate between tumour and non-tumor instances, these pre-trained networks are adjusted.

To guarantee compliance with the model input pipeline, MRI images from the TCGA dataset are cleaned, scaled to 256 × 256, and formatted during the preprocessing step. After then, the dataset is divided in a 72:20:8 ratio into training, validation, and testing sets. To input these photos in mini-batches, normalise them, and create category target masks, a data generator is set up.

In the classification phase, each input MRI scan $X \in R^{256\times256\times C}$ is passed through a pre-trained CNN base. Feature maps F(X) are extracted and processed through a global average pooling layer followed by a dense classification head. The output $\hat{y} = \sigma(WF(X) + b)$ represents the probability of tumor presence, with training guided by a binary cross-entropy loss function:

$$L_{BCE} = -[y \log(\hat{y}) + (1-y)\log(1-\hat{y})]$$

where y is the ground truth label (1 for tumor, 0 for no tumor). If a tumor is detected ($\hat{y} > 0.5$), the image is then passed to the segmentation phase.

The segmentation model is based on the RESUNet architecture, which is an enhanced version of the U-Net with integrated residual connections. Each residual block includes two convolutional layers, batch normalization, ReLU activation, and an identity skip connection. Mathematically, each residual block is represented as:

$$ResBlock(x) = x + f(x)$$

where f(x) denotes the non-linear transformation applied to the input x. The encoder path captures high-level features while progressively downsampling the spatial dimensions, and the decoder path upsamples these features to reconstruct the segmentation mask. Feature maps from the encoder are concatenated with decoder outputs at corresponding levels to preserve spatial context.

The final segmentation output, denoted as $\hat{y}M(i, j)$, lies in the range [0, 1] and is obtained using a sigmoid-activated 1 × 1 convolution, indicating the per-pixel probability of tumor presence.

Training this model uses a composite loss function that combines binary cross-entropy loss and Dice loss to balance pixel-wise accuracy with overlap-based metrics:

$$L_{seg} = \alpha L_{BCE} + (1-\alpha)L_{Dice}$$

With Dice loss defined as:

$$L_{Dice}1 - \frac{2\sum M(i,j)\widehat{M}(i,j)}{\sum M(i,j)^2 + \sum \widehat{M}(i,j)^2}$$

where M (i, j) is the ground truth mask. The result is a high-resolution segmentation map highlighting tumor regions, overlaid on the MRI for clinical interpretation (see Table 1).

This overall framework [Fig. 8] ensures accurate tumor classification and localization, combining the coarse-level judgment of CNN-based classification with the fine-grained prediction power of RESUNet, making it suitable for real-time clinical applications in radiology and oncology. Table 2 is presenting the hyperparameter and training configuration for the proposed work.

The unique aspects of our proposed pipeline—specifically, the dual-

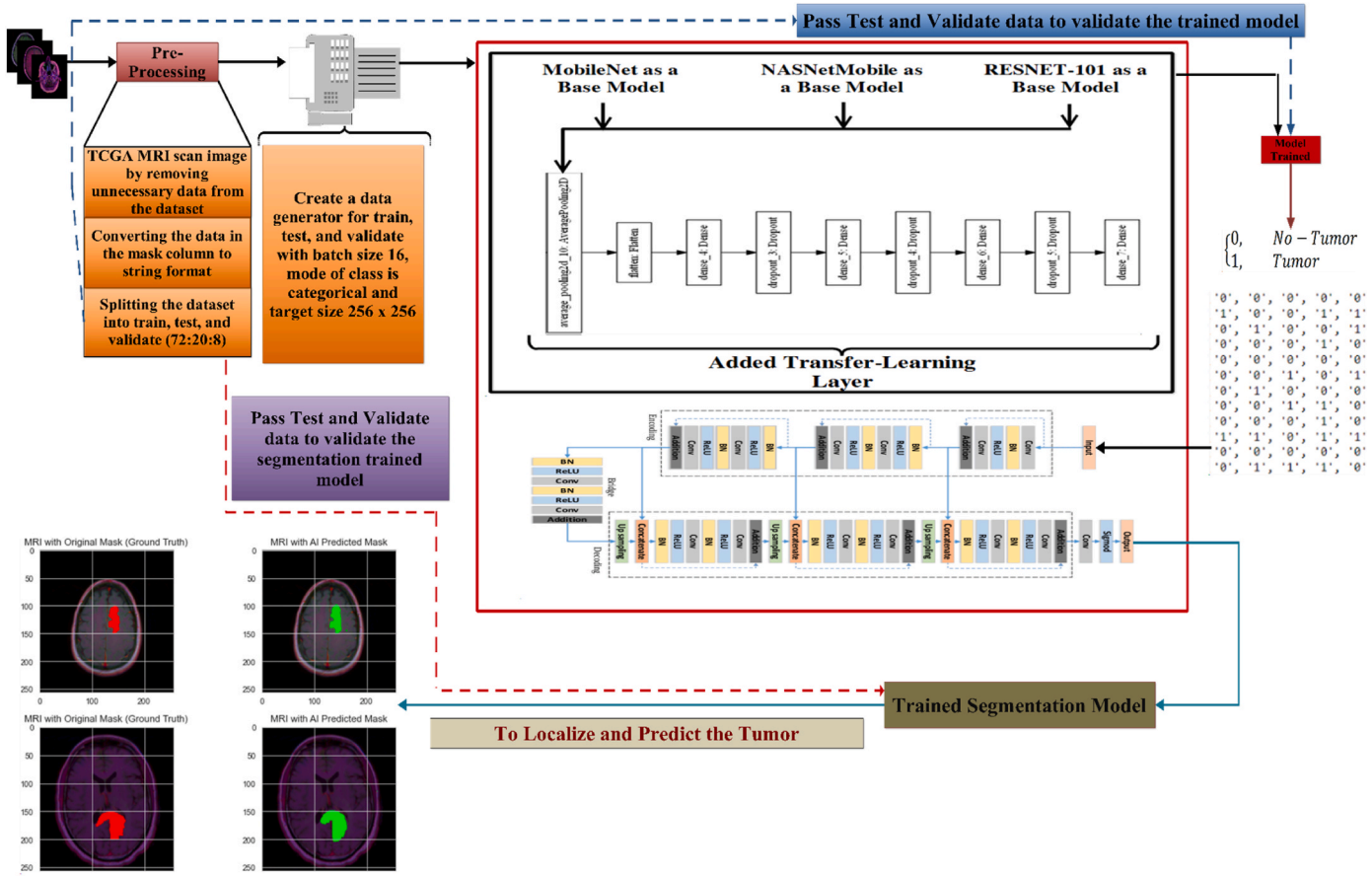


Fig. 8. Overall framework.

Table 1
Related work.

Ref	Thematic Category	Model Type	Innovation	Notable Strength	Research Gap
[11]	Hybrid (CNN + Attention)	Global-Local Parallel Dual-Branch CNN	Attention-enhanced feature fusion between global and local contexts	Captures both local tumor texture and global contextual cues	High model complexity; needs optimization for real-time use
[12]	CNN-based	Lightweight Multi-Path CNN	Optimal feature selection for efficient multiclass classification	Low computational cost; suitable for embedded systems	May underperform on large-scale, diverse MRI datasets
[13]	Hybrid (CNN + MAMBA)	Parallel CNN-MAMBA	Integration of CNN and MAMBA for parallel learning	Enhanced feature extraction and robustness	Limited explainability of MAMBA integration
[14]	Hybrid (Ensemble CNN)	Xception + Parallel CNN Ensemble	Combines multiple CNNs for improved generalization	High accuracy through model diversity	Computationally expensive for training
[15]	CNN-based	SAlexNet (Residual Attention)	Incorporates residual attention in AlexNet for fine-grained detection	Efficient and accurate for primary tumor detection	Limited evaluation on multi-class tumor subtypes
[16]	Transformer-based	DieT Transformer with PCA-ADE	Combines PCA and ADE with Vision Transformer	Superior performance on multi-class tasks	Requires high computational resources
[17]	CNN-based	Deep Learning with GDD Feature Approximation	Novel GDD-based feature extraction for classification and survival prediction	Integrates classification with survival analysis	Lack of external dataset validation
[18]	Hybrid (Ensemble CNN)	Two-Stage CNN Ensemble	Score-level fusion of multiple CNNs	Improved robustness against noise and variability	Requires tuning for class imbalance handling
[19]	CNN-based	Early Detection CNN	Early-stage tumor detection using deep learning	Demonstrates clinical potential in early diagnosis	Needs validation with real clinical datasets
[20]	CNN-based	Data-Augmented CNN	Simulation of real-life scenarios for robust classification	Strong generalization through augmentation	Over-reliance on synthetic data quality
[21]	Hybrid (ML + Transfer Learning)	Ensemble of ML and Transfer Learning Models	Comparative framework of traditional and deep models	Broad coverage of learning paradigms	Limited model interpretability
[22]	Hybrid (Multi-Task DL)	BT-Net (Classification + Segmentation + Localization)	Unified model for multi-task MRI analysis	Reduces pipeline fragmentation	High training cost and resource demand
[23]	CNN-based	Fine-tuned EfficientNet	EfficientNet fine-tuning for multigrade classification	Strong feature reuse and efficiency	Dataset-specific tuning required
[24]	CNN-based	Enhanced MRI CNN	Performance optimization using MRI-specific features	Consistent improvements across metrics	Limited novelty in architecture
[25]	Hybrid (Feature-based DL)	Modified DL with Hybrid Feature Extraction	Combines deep and handcrafted features	Improves interpretability and accuracy	Complexity increases training time
[26]	CNN-based (Transfer Learning)	Fine-Tuned Pretrained CNNs	Transfer learning with fine-tuning strategy	Effective even with limited labeled data	Dependent on source domain similarity
[27]	CNN-based (IoMT)	Deep Learning on IoMT	Deploys CNN classification on IoMT frameworks	Enables remote and continuous diagnosis	Network security and latency concerns
[28]	Hybrid (Segmentation + Classification)	EMU-Net (Modified U-Net)	Combines segmentation and classification in U-Net	Improved localization and accuracy	High resource demand for real-time deployment
[29]	Hybrid (CNN + ELM)	RBEBT (ResNet + BA-ELM)	Merges CNN feature extraction with ELM classifier	Fast classification and strong performance	Limited scalability on larger datasets
[30]	CNN-based (Data Augmentation)	CNN with Synthetic Data	Uses synthetic data to enhance model generalization	Mitigates data scarcity	Synthetic data realism affects accuracy
[31]	Hybrid (SVM + Texture Features)	LS-SVM with Multi-Model Texture Features	Combines handcrafted texture descriptors with SVM	High accuracy and interpretability	Not end-to-end trainable; limited deep feature use
[32]	Hybrid (Ensemble Learning)	Multi-Model Ensemble	Combines diverse learners (CNN + ML) for MRI classification	Improved stability and performance	Model fusion may increase computational load
[33]	Hybrid (Ensemble DL)	Weighted Average Ensemble DL	Weighted fusion of multiple DL models	Enhances reliability and precision	Limited adaptability to unseen data distributions

stage integration where transfer learning-based CNN classifiers (MobileNet, NASNetMobile, ResNet101) first identify tumor presence, and the output guides a RESUNET segmentation network for precise localization. This sequential coupling allows feature refinement between stages, improving detection accuracy and segmentation precision.

4. Experiment and Analysis

4.1. Datasets and visualisation

This research utilized the Cancer Genome Atlas Glioblastoma Multiforme (TCGA-GBM) dataset [43], which is available through The Cancer Imaging Archive (TCIA), to assess the performance of a proposed deep learning framework that integrates a Base model with a RESUNET architecture. The TCGA-GBM dataset, a subset of the broader TCGA project, specifically focuses on Glioblastoma Multiforme—an aggressive form of brain cancer. While the TCGA project includes roughly 3929 imaging samples across various cancer types, this study focused on brain tumor MRI scans. The Python pandas module was used to read the CSV file containing the necessary information that goes with these photos. Four important columns—patient_id, image_path, mask_path, and mask—are included in the brain_dataframe pandas DataFrame, which was loaded with the dataset. Because it provides binary values that indicate whether a tumour is present in the accompanying MRI picture (usually, 0 for no tumour and 1 for tumour existence), the mask column is very significant. Smooth preprocessing and model training are made possible by this organised data format.

To aid in visualizing the class distribution of the dataset, a bar chart was developed [Fig. 9]. Counting the values in the mask column helps the experiment create a bar graph showing the frequency of every class—tumor against non-tumor. Building reliable machine learning models depends on this visualisation displaying the balance or imbalance of the dataset. If the data is highly biassed towards one class, further preprocessing techniques like class weighting or oversampling might be needed to assure excellent model performance.

The tumour localization capacity of the model was experimentally tested using many MRI sequences; T1-weighted post-contrast images enhanced with gadolinium (T1-Gd) showed the greatest accuracy. Usually in tumors, the gadolinium agent accumulates in areas where the blood-brain barrier is disrupted and offers better contrast between cancer and healthy tissue on these scans. Thanks to this contrast enhancement, the deep learning model—especially the RESUNET

Table 2
Hyperparameter and training Configuration.

Parameter	Description/Setting	Value/Configuration
Optimizer	Optimization algorithm used for model training	Adam
Loss Function	Objective function for multi-class classification	Categorical Crossentropy
Metrics	Performance evaluation metric	Accuracy
Epochs	Maximum number of training iterations	100
Batch Size	Number of samples per gradient update	16
Early Stopping (monitor)	Performance metric monitored for early stopping	val_loss
Early Stopping (mode)	Optimization direction for monitored metric	min
Early Stopping (patience)	Epochs with no improvement before stopping	25
ReduceLROnPlateau (monitor)	Metric monitored for learning rate reduction	val_loss
ReduceLROnPlateau (mode)	Optimization direction for monitored metric	min
ReduceLROnPlateau (patience)	Epochs with no improvement before reducing LR	10
ReduceLROnPlateau (factor)	Factor by which the learning rate is reduced	0.2
ReduceLROnPlateau (min_delta)	Minimum change to qualify as improvement	0.0001
Model Checkpoint (monitor)	Metric monitored for saving best model	val_loss
Model Checkpoint (save_best_only)	Saves only model with best validation loss	True
Callbacks Used	List of active callbacks during training	[ModelCheckpoint, EarlyStopping, ReduceLROnPlateau]

component—can more accurately identify and separate tumors boundaries. Thanks to the integration of the TCGA-GBM dataset, organised metadata processing using Pandas, class display using Plotly, and T1-Gd imaging, the proposed tumour detection model was successfully implemented and assessed overall.

Confirming the quality of the TCGA-GBM dataset required this basic visualisation [Fig. 10] step prior to model training. Presenting pairs of MRIs and the associated tumors, the researchers ensured the data was suitably tagged and fit for deep learning segmentation. Furthermore, this phase helped the model building process by offering perceptive analysis on tumour variability.

The research proceeded with an enhanced visualisation technique [Fig. 11] as the following step in data exploration to help grasp of the MRI data and the related tumors masks. This was accomplished by building a composite view wherein the raw MRI scan, mask, and integrated image from each instance could be shown side by side. The brain_dataframe contains binary masks and paths to the MRI pictures, hence the program selected five samples at random from there. For better handling of pixel arrays, this was achieved by simply putting the mask straight onto the MRI image using NumPy indexing once the images were read using skimage.io.imread. Specifically, the line img[mask == 255] = (255, 0, 0) modified the MRI's pixel values where the mask indicated the presence of a tumour (value 255), therefore rendering those regions brilliant red. Thanks to the rapid visual identification of cancer boundaries within the MRI environment, tumour location, shape, and size may now be more easily comprehended. Medical imaging uses this method of superimposing the mask over the original MRI as it resembles how a radiologist may see a picture with comments. It enables quick data integrity checking and aesthetically confirms proper alignment of the mask with anatomical characteristics. Furthermore, it helps interpretability when evaluating segmentation models by providing a direct comparison of models.

4.2. Data preprocessing and generator setup

The preprocessing pipeline implemented in this research was designed to optimize the TCGA-GBM brain MRI dataset for effective use in a supervised deep learning classification framework. The steps were carefully structured to eliminate redundancy, enforce data compatibility with neural network requirements, and preserve the integrity of class distributions during training and evaluation.

4.2.1. Metadata refinement and DataFrame construction

Metadata like patient_id, image_path, mask_path, and mask were present in the original dataset. Although it provided a unique identification, the patient_id column had no predictive data pertinent to the learning job. Such non-informative characteristics may cause the model to become noisy or too complicated. Thus, Pandas' drop function was used to eliminate this column. Only the following crucial elements were kept in the final DataFrame, brain_dataframe_train:

- image_path: File path to the MRI image.
- mask_path: File path to the corresponding ground truth mask image.
- mask: A binary label indicating the presence (1) or absence (0) of tumor tissue in the scan.

This refined structure ensured a clean and focused dataset aligned with the supervised classification objective.

4.2.2. Label formatting for categorical modelling

The mask column, initially stored as an integer (0 or 1), was

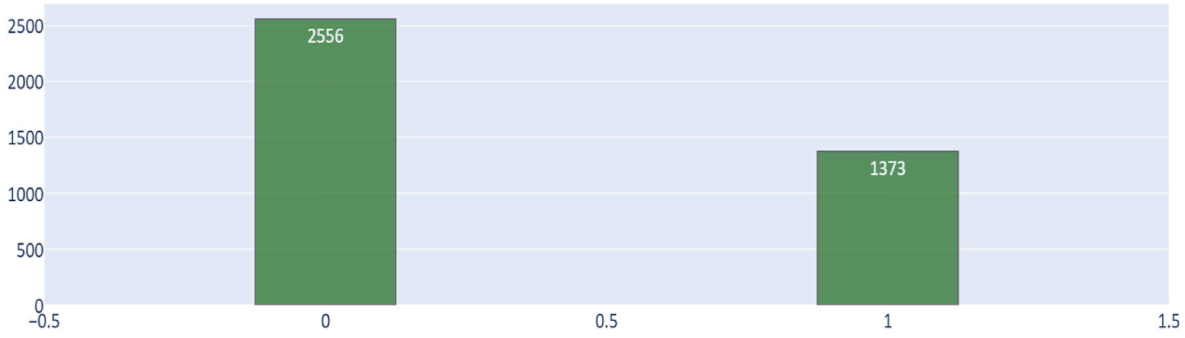


Fig. 9. Distribution of tumor and non tumor images.

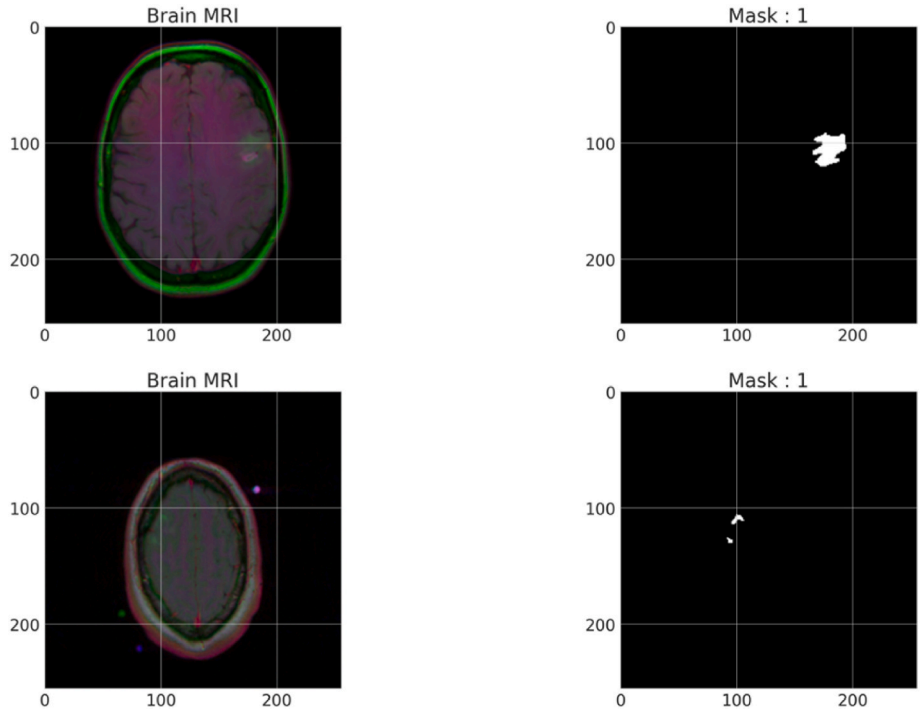


Fig. 10. Basic visualisation.

converted to a string. This transformation was essential because Keras ImageDataGenerator expects labels to be in string format when the class_mode = 'categorical' parameter is used. Internally, this mode performs one-hot encoding, converting binary classes into vectors like [1, 0] and [0, 1], enabling compatibility with softmax activation in the model's final layer. This conversion makes the dataset readily useable for training multi-class or binary classifiers.

4.2.3. Data partitioning strategy

To evaluate model performance effectively, the dataset was split into training and testing subsets using the train_test_split function from the scikit-learn library. A split ratio of 80:20 was applied—80% for training and 20% for testing—ensuring a large enough dataset for model learning while reserving a representative hold-out set for unbiased performance evaluation.

Furthermore, 10% of the training data was internally designated as a validation set using the validation_split = 0.1 argument in the Keras ImageDataGenerator [Fig. 12]. This resulted in:

- 72% training data (i.e., 90% of the 80%),
- 8% validation data, and
- 20% testing data.

This structured partitioning supports three core objectives:

- Enables the model to generalise by learning from the majority of the dataset,
- Monitors validation loss and accuracy during training to mitigate overfitting,
- Provides a test set for final evaluation, untouched by any part of the training pipeline.

4.2.4. Normalization and generator initialization

To prepare image inputs for the deep learning model, all MRI scans and mask images were rescaled using the formula rescale = 1./255. This normalization transforms pixel intensity values from a range of 0–255 (standard in 8-bit grayscale or RGB images) to a range of 0–1, which is numerically more stable for neural networks and accelerates convergence during backpropagation.

The ImageDataGenerator class in Keras was employed to handle data loading, preprocessing, and augmentation in memory-efficient batches. The generators were configured as follows:

4.2.4.1. Training generator.

- Pulled from the training portion of the DataFrame.
- Set subset = 'training' to indicate it's part of the training split.
- Enabled shuffling to promote learning across varied examples.
- Used class_mode = 'categorical' to handle binary labels as categorical data.

4.2.4.2. Validation generator.

- Extracted from the same DataFrame with subset = 'validation'.
- Used the same preprocessing as the training generator.
- Helped monitor validation loss and accuracy during model training for hyperparameter tuning and early stopping.

4.2.4.3. Test generator.

- Constructed using a separate ImageDataGenerator, configured only with rescale = 1./255.
- No augmentation or shuffling was applied (shuffle = False) to maintain data order, which is critical for associating predicted labels with actual ground truth during post-hoc analysis and metric computation.

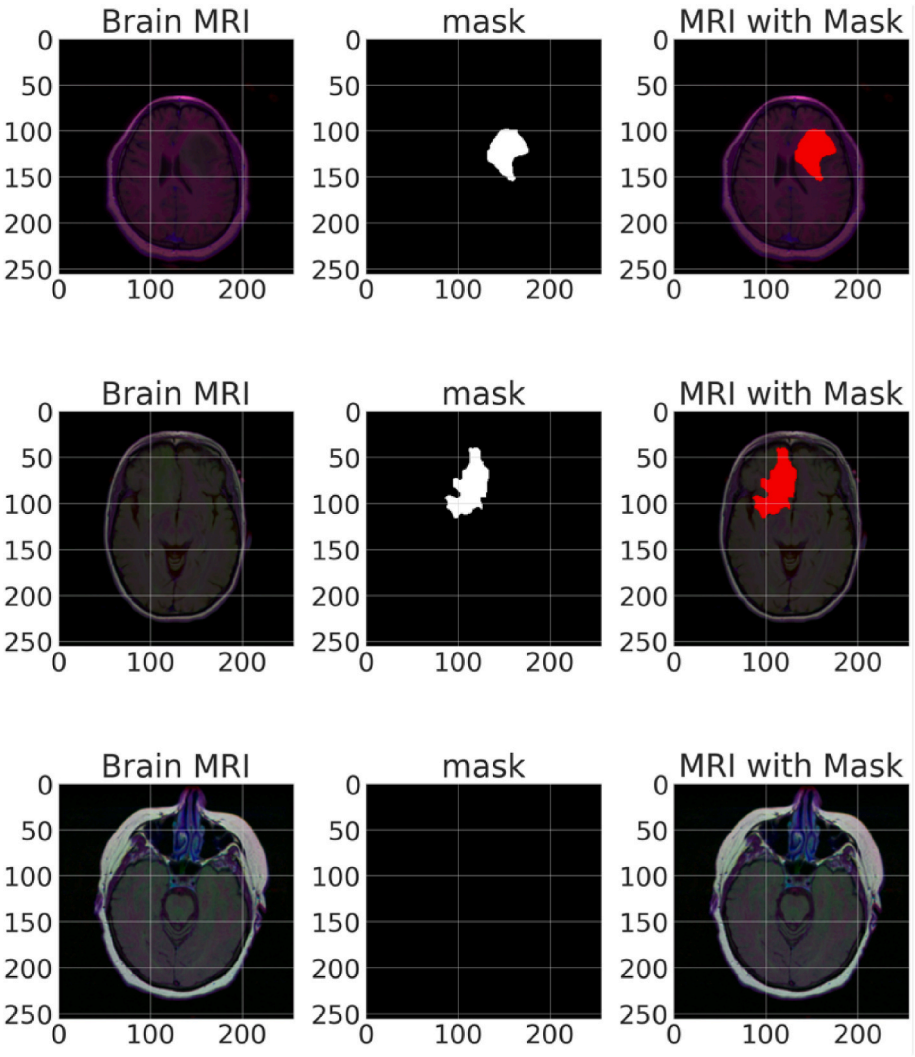


Fig. 11. Advance visualisation.

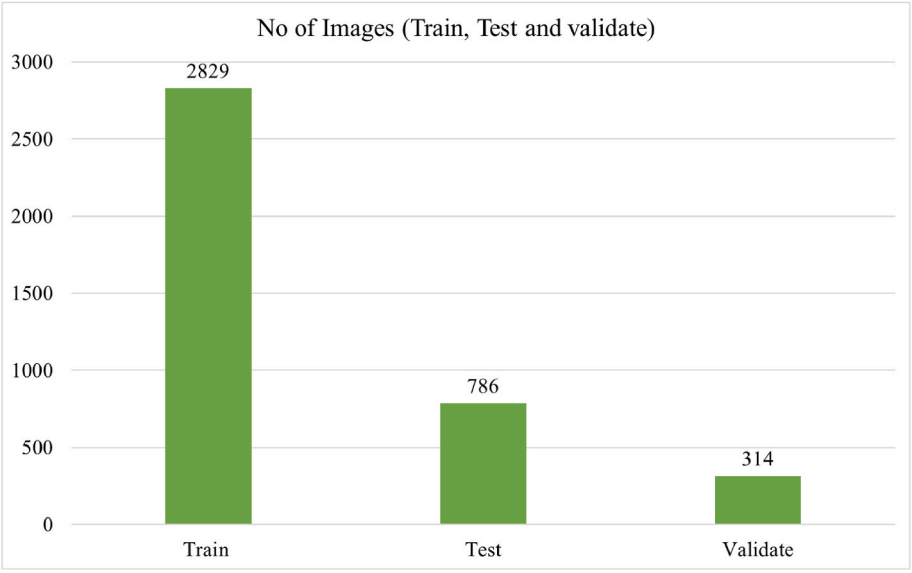


Fig. 12. Number of images in train, test and validate.

All generators resized the images to a standardized shape of 256 × 256 pixels (target_size=(256,256)), ensuring uniform input dimensions

for the convolutional neural network. A batch size of 16 was selected to balance memory efficiency with learning stability.

4.3. Evaluation matrics

4.3.1. Accuracy - the generalist lens

Purpose: Captures overall classification correctness across all categories.

Insight: Accuracy provides a broad overview of model performance by quantifying the ratio of correct predictions to total predictions. It is most reliable in balanced datasets but may offer inflated impressions in skewed distributions.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

where:

- TP: True Positives — the number of positive instances correctly identified by the model.
- TN: True Negatives — the number of negative instances correctly identified by the model.
- FP: False Positives — the number of negative instances incorrectly identified as positive by the model.
- FN: False Negatives — the number of positive instances incorrectly identified as negative by the model.

Role: Acts as a baseline metric — a macro-level indicator of correctness.

4.3.2. Precision – the certainty lens

Purpose: Measures the reliability of positive predictions.

Insight: Precision is sensitive to false positives, making it particularly valuable when the cost of incorrect positive predictions is high (e.g., in disease diagnosis or spam detection). It answers the question: "When the model predicts positive, how often is it correct?"

$$Precision = \frac{TP}{TP + FP}$$

Role: Highlights the model's capacity to avoid overgeneralizing positives.

4.3.3. Recall - the completeness lens

Purpose: Evaluates the model's ability to capture all true positives.

Insight: Recall, or sensitivity, is critical when missing a true positive carries significant consequence (e.g., failure to detect cancer in medical images). It assesses how comprehensively the model identifies the positive class.

$$Recall = \frac{TP}{TP + FN}$$

Role: Measures coverage of relevant instances, especially important in high-risk domains.

4.3.4. F1 score – the balance lens

Purpose: Harmonizes precision and recall into a unified metric.

Insight: The F1 score is the harmonic mean of precision and recall, offering a single scalar value that penalizes extreme imbalances between the two. It is especially useful in domains where both types of classification errors carry weight.

$$F1 = \frac{2 \cdot TP}{2 \cdot TP + FP + FN}$$

Role: Acts as a compromise metric—useful when neither false positives nor false negatives can be ignored.

4.3.5. Tversky Loss –- the customizable discrimination lens

Purpose: Allows flexible weighting of false positives and false negatives.

Insight: Tversky loss is particularly suited for image segmentation and imbalanced datasets. It generalizes the Dice coefficient by introducing tunable parameters (α, β) that control the penalty for each type of error. This adaptability makes it ideal for domain-specific error prioritization.

$$Tversky\ Index = \frac{|A \cap B|}{|A \cap B| + \alpha|A - B| + \beta|B - A|}$$

where:

- A: The set of true positive pixels (actual positives).
- B: The set of predicted positive pixels.
- α: Weighting factor that penalizes false positives.
- β: Weighting factor that penalizes false negatives.

Role: A domain-sensitive metric tailored for tasks requiring nuanced performance evaluation.

5. Result

Table 3 and Figs. 13–15 illustrate the accuracy performance of the proposed network on the given dataset. Similarly, Table 4 and Figs. 16, 18 and 20 depict the loss performance, while Table 5 and Figs. 17, 19 and 21 present the classification report and epoch-wise Tversky loss performance.

In order to assess classification models, the confusion matrix [Fig. 22] maps predicted vs real labels into four basic categories: True Positives (TP), False Positives (FP), True Negatives (TN), and False Negatives (FN). Every element represents a distinct model decision behaviour. While True Negatives relate to the accurate identification of negative cases, True Positives happen when the model properly detects positive examples. False Positives, also known as Type I mistakes, are situations in which the model misclassifies a negative sample as positive. These are particularly important in applications such as spam detection or medical screenings, when false alarms may lead to needless worry or action. In delicate fields like cancer diagnosis or fraud detection, false negatives, also known as Type II mistakes, may be very harmful when positive cases are mistakenly categorised as negative.

Choosing the right loss function is essential to model efficacy in medical image classification, especially tumour identification, since class imbalance is a prevalent feature of these datasets (e.g., less tumour pixels vs. healthy tissue). Because of their strong convergence characteristics and ease of mathematics, traditional loss functions like categorical cross-entropy are often used. But since they assume a balanced class distribution and provide equal attention to each class, they often favour the dominant class, which leads to unsatisfactory performance for minority classes (such as tumour areas). This may result in low tumour detection sensitivity but high overall accuracy, which is an undesirable trade-off in clinical situations.

On the other hand, the Tversky Loss, derived from the Tversky Index, addresses this shortcoming by introducing two tunable parameters, α

Table 3
Deep Learning Models with Transfer Layer and their Average Accuracy.

S No	Deep Learning Model with Transfer Layer	Average Accuracy	Trainable Parameter in Deep Learning Framework
1	MobileNet + Transfer Learning Layer	0.9600	1,180,930
2	NASNetMobile + Transfer Learning Layer	0.9700	5446676
3	ResNet101 + Transfer Learning Layer	0.9500	44782338

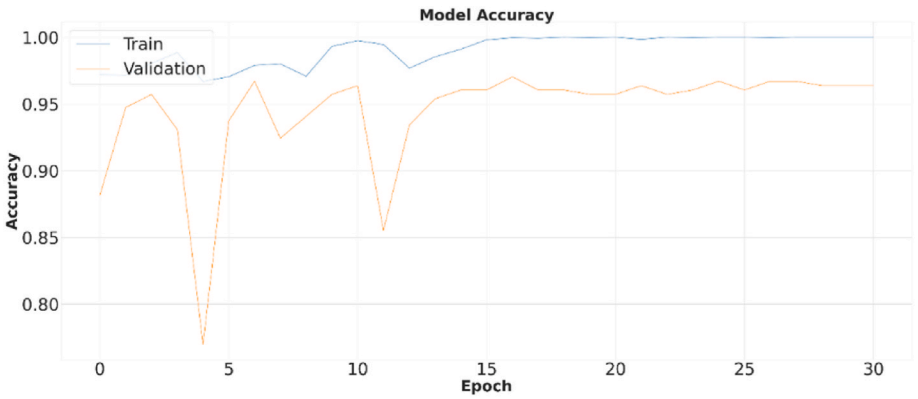


Fig. 13. Epoch by epoch accuracy MobileNet-Transfer learning layer.

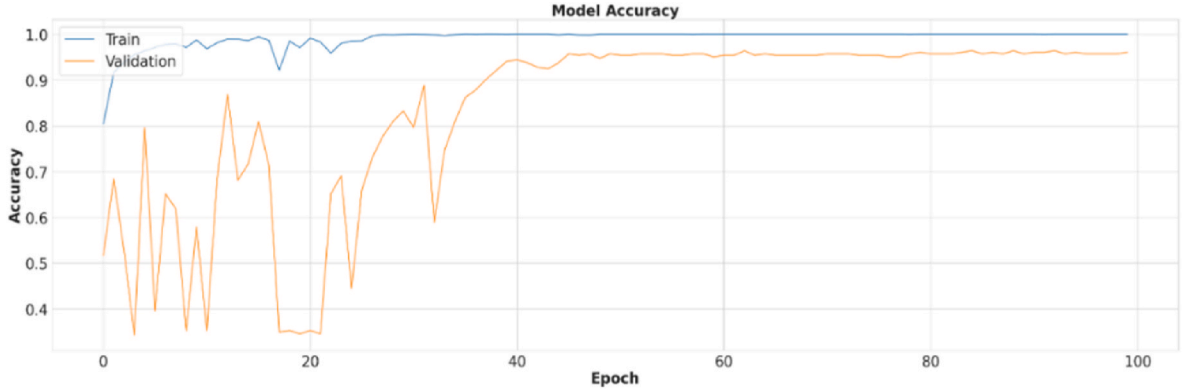


Fig. 14. Epoch by epoch accuracy NASNetMobile-Transfer learning layer.

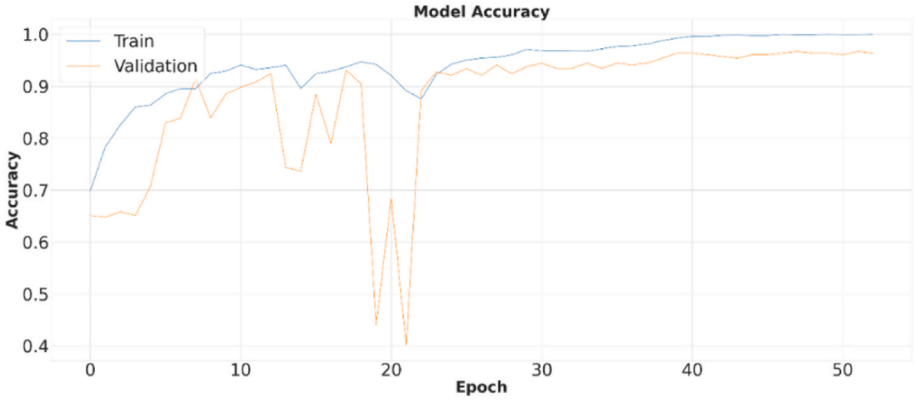


Fig. 15. Epoch by epoch accuracy ResNet101-Transfer learning layer.

Table 4
Loss and Tversky loss performance for classification model.

S No	Deep Learning Model with Transfer Layer	Loss	Tversky Loss
1	MobileNet + Transfer Learning Layer	0.1556	0.9161
2	NASNetMobile + Transfer Learning Layer	0.1580	0.9142
3	ResNet101 + Transfer Learning Layer	0.1659	0.9085

and β, that respectively control the penalty for false positives and false negatives. This tunability is especially useful in tumor classification where false negatives (i.e., missing an actual tumor) are significantly more critical than false positives. By emphasizing recall over precision, the Tversky Loss skews the model's learning towards capturing all positive (tumor) cases—even at the expense of a few false alarms—thus aligning better with clinical priorities where early detection outweighs over-detection. Furthermore, the Tversky Loss generalizes the Dice Loss, allowing for finer control over class imbalance and facilitating better localization of small and irregular tumor shapes.

Hence, while cross-entropy or general loss functions may suffice for balanced tasks, Tversky Loss is particularly suited to tumor classification problems due to its sensitivity to imbalance, its customizable penalty structure, and its proven superiority in segmentation and classification tasks where minority class accuracy is paramount. In critical health-related domains, prioritizing true positive recognition justifies the adoption of such domain-sensitive loss functions.

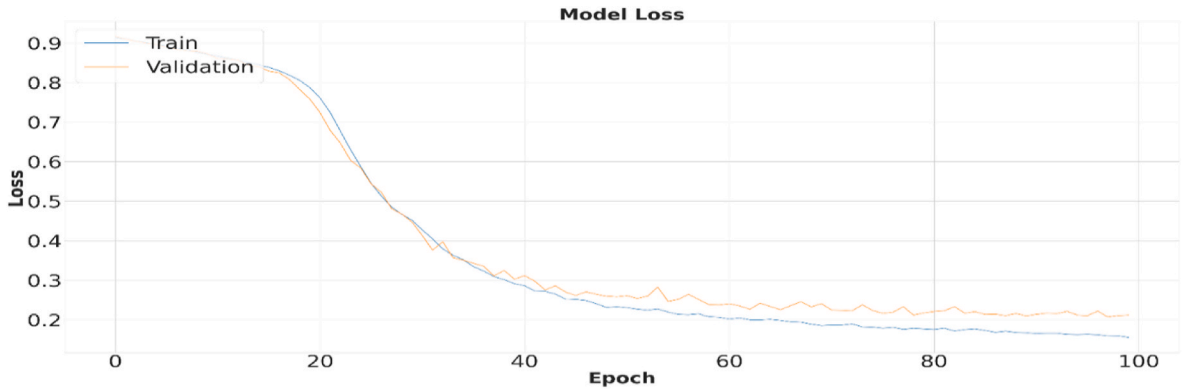


Fig. 16. Epoch by epoch loss graph for MobileNet-Transfer learning.

Table 5
Classification report.

S No	Deep Learning Model with Transfer Learning Layer	Parameter	Precision	Recall	F1 Score
1	MobileNet + Transfer Learning Layer	0	0.95	0.99	0.97
		1	0.98	0.92	0.95
		Accuracy	0.96	0.96	0.96
		Micro Average	0.97	0.95	0.96
		Weighted Average	0.96	0.96	0.96
2	NASNetMobile + Transfer Learning Layer	0	0.96	0.99	0.98
		1	0.99	0.93	0.96
		Accuracy	0.97	0.97	0.97
		Micro Average	0.97	0.96	0.97
		Weighted Average	0.97	0.97	0.97
3	ResNet101 + Transfer Learning Layer	0	0.97	0.96	0.96
		1	0.93	0.94	0.94
		Accuracy	0.95	0.95	0.95
		Micro Average	0.95	0.95	0.95
		Weighted Average	0.95	0.95	0.95

6. Discussion

Figs. 23–25 illustrate the localization results of the proposed framework using three different backbone configurations:

1. MobileNet integrated with a Transfer Learning Layer and the RESUNET model,
2. NASNetMobile integrated with a Transfer Learning Layer and the RESUNET model, and
3. ResNet101 integrated with a Transfer Learning Layer and the RESUNET model.

A comparative analysis of various deep learning models demonstrates a clear advancement in performance from traditional CNN-based techniques to more sophisticated architectures that leverage hybrid methods and transfer learning. Earlier models such as CNN-GAN (accuracy: 0.8653), CNN-KNN (0.8750), and CNN-SVM (0.9310) show moderate effectiveness, with CNN-SVM performing best among them. These methods typically depend on handcrafted feature extraction followed by classical machine learning classifiers, which limits their capacity to learn complex data representations. More advanced deep learning architectures, including DenseNet-121 (0.9210), GoogLeNet (0.8730), MobileNet (0.8890), and ResNet-50 (0.9530), show notable improvements. Among them, ResNet-50 stands out for its use of deep residual connections, which enhance representation learning. Additionally, segmentation-oriented models like UNet with ResNeXt50 (0.9320) and UNet-DCNN (0.8860) demonstrate how encoder-decoder structures, combined with powerful feature extractors, can significantly boost performance in image classification and segmentation tasks. Hybrid and ensemble models further elevate performance levels. For example, QDCNN-DMN (0.9400) benefits from combining multiple models or learning stages to extract a richer feature set. Remarkably, the MSMCNN-LSTM model achieves an accuracy of 0.9636 by leveraging the temporal modeling capabilities of LSTM networks, which capture dependencies across spatial features. The most promising outcomes are observed in the proposed models employing transfer learning. MobileNet + Transfer Learning (0.9600), ResNet101 + Transfer Learning (0.9500), and especially NASNetMobile + Transfer Learning (0.9700) surpass other approaches. These results highlight the strength of

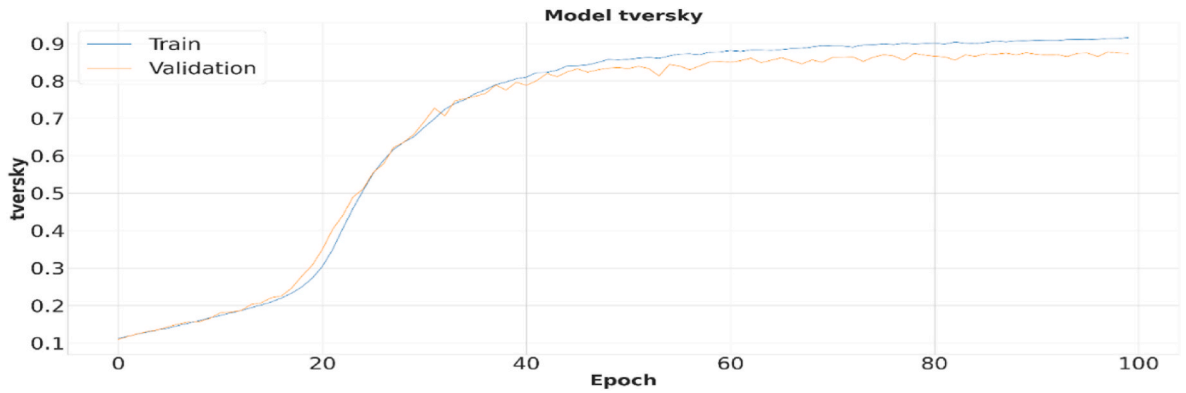


Fig. 17. Epoch by epoch Tversky graph for MobileNet-Transfer learning.



Fig. 18. Epoch by epoch loss graph for NASNetMobile-Transfer learning.

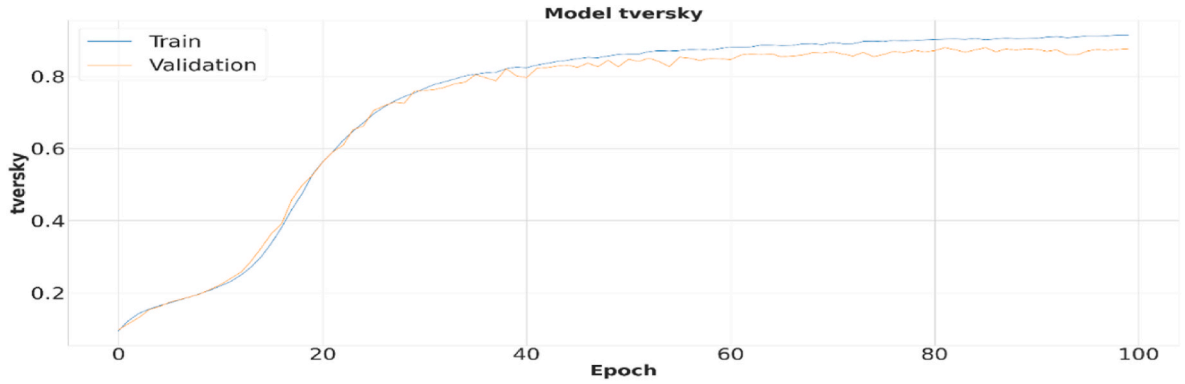


Fig. 19. Epoch by epoch Tversky graph for NASNetMobile-Transfer learning.

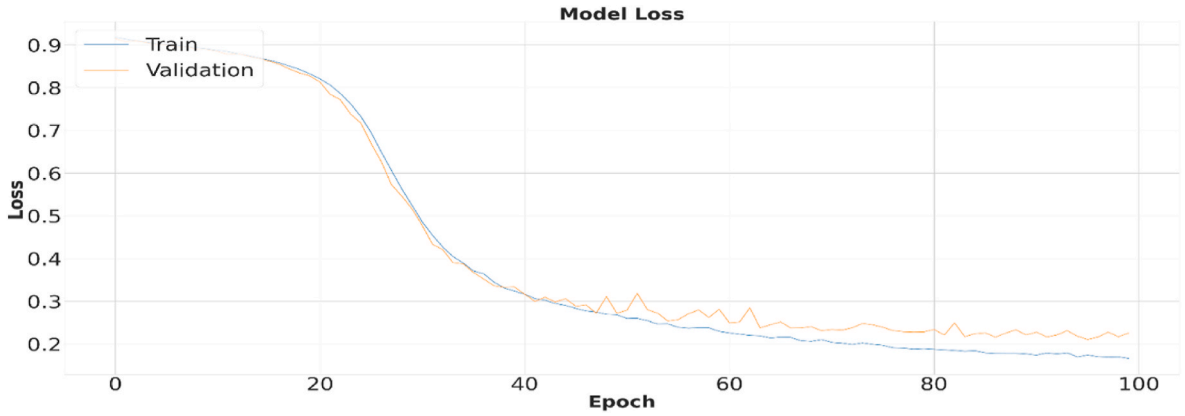


Fig. 20. Epoch by epoch loss graph for ResNet101-Transfer learning.

utilizing pre-trained models that are fine-tuned on domain-specific data. NASNetMobile, in particular, achieves the highest accuracy, demonstrating that lightweight architectures, when paired with transfer learning, can deliver exceptional performance.

Overall, the comparison shows a clear trend: contemporary deep learning methods—especially those including hybrid architectures and transfer learning—offer significant advantages over conventional approaches. As deep networks grow into more specialised and data-efficient systems, their ability to generalise and perform effectively over demanding tasks is improving. Table 6 provides a detailed evaluation of numerous innovative techniques.

Looking at the three confusion matrices [Fig. 22] for MobileNet + Transfer Learning, NASNetMobile + Transfer Learning, and ResNet101 + Transfer Learning, study can analyze the failure cases (misclassifications) in the context of brain tumor identification from MRI images [Table 7].

6.1. *False Positives (FP)*

These are non-tumor MRIs predicted as tumor. In a clinical setting, this could lead to unnecessary follow-up tests or anxiety for patients.

- **MobileNet + Transfer Learning Layer**: 6 FP
- **NASNetMobile + Transfer Learning Layer**: 4 FP (best among the three)

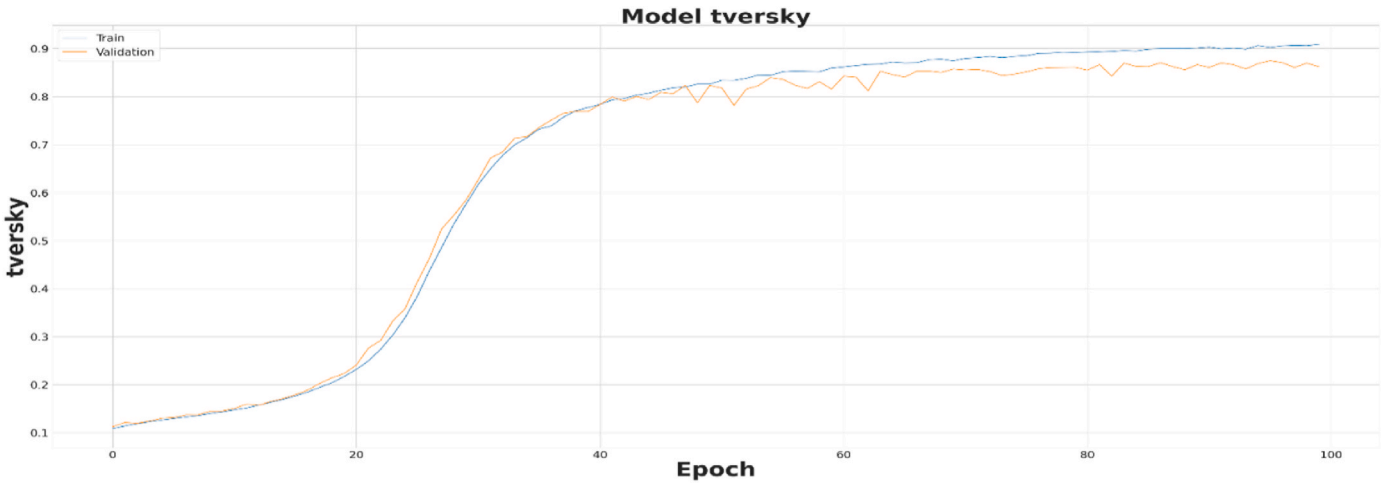


Fig. 21. Epoch by epoch Tversky graph for ResNet101-Transfer learning.

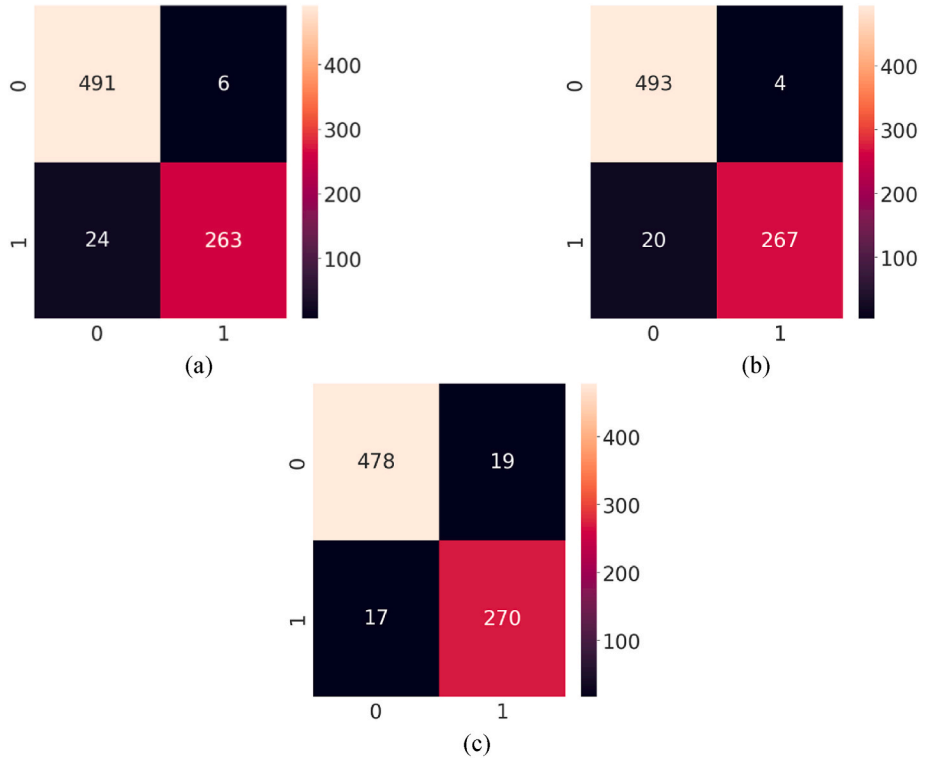


Fig. 22. Confusion matrix for MobileNet + transfer learning layer, NASNetMobile + transfer learning layer, ResNet101 + transfer learning layer.

- **ResNet101 + Transfer Learning Layer**: 19 FP (highest, worst-case scenario)

Observation: NASNetMobile minimizes overdiagnosis, while ResNet101 tends to over-predict tumors more often.

6.2. *False Negatives (FN)*

These are tumor MRIs predicted as non-tumor, which is more critical because a tumor could be missed.

- **MobileNet + Transfer Learning Layer**: 24 FN
- **NASNetMobile + Transfer Learning Layer**: 20 FN
- **ResNet101 + Transfer Learning Layer**: 17 FN (best performance in detecting tumors)

Observation: ResNet101 is more sensitive to tumors, reducing the risk of missing actual tumor cases. MobileNet misses slightly more tumors, which could be clinically significant.

MobileNet + Transfer Learning Layer demonstrates balanced performance, with the advantage of a low number of false positives, which helps avoid unnecessary interventions. However, it has a slightly higher number of false negatives compared to ResNet101 + Transfer Learning Layer, meaning it may miss more tumors—a clinically significant risk. Overall, MobileNet + Transfer Learning Layer is relatively safe, but careful follow-up may be needed to ensure missed tumors are detected. NASNetMobile + Transfer Learning Layer excels at minimizing false positives, making it safer for avoiding unnecessary anxiety or procedures, though it is slightly less sensitive than ResNet101 + Transfer Learning Layer in detecting tumors. This makes NASNetMobile + Transfer Learning Layer particularly suitable in contexts where

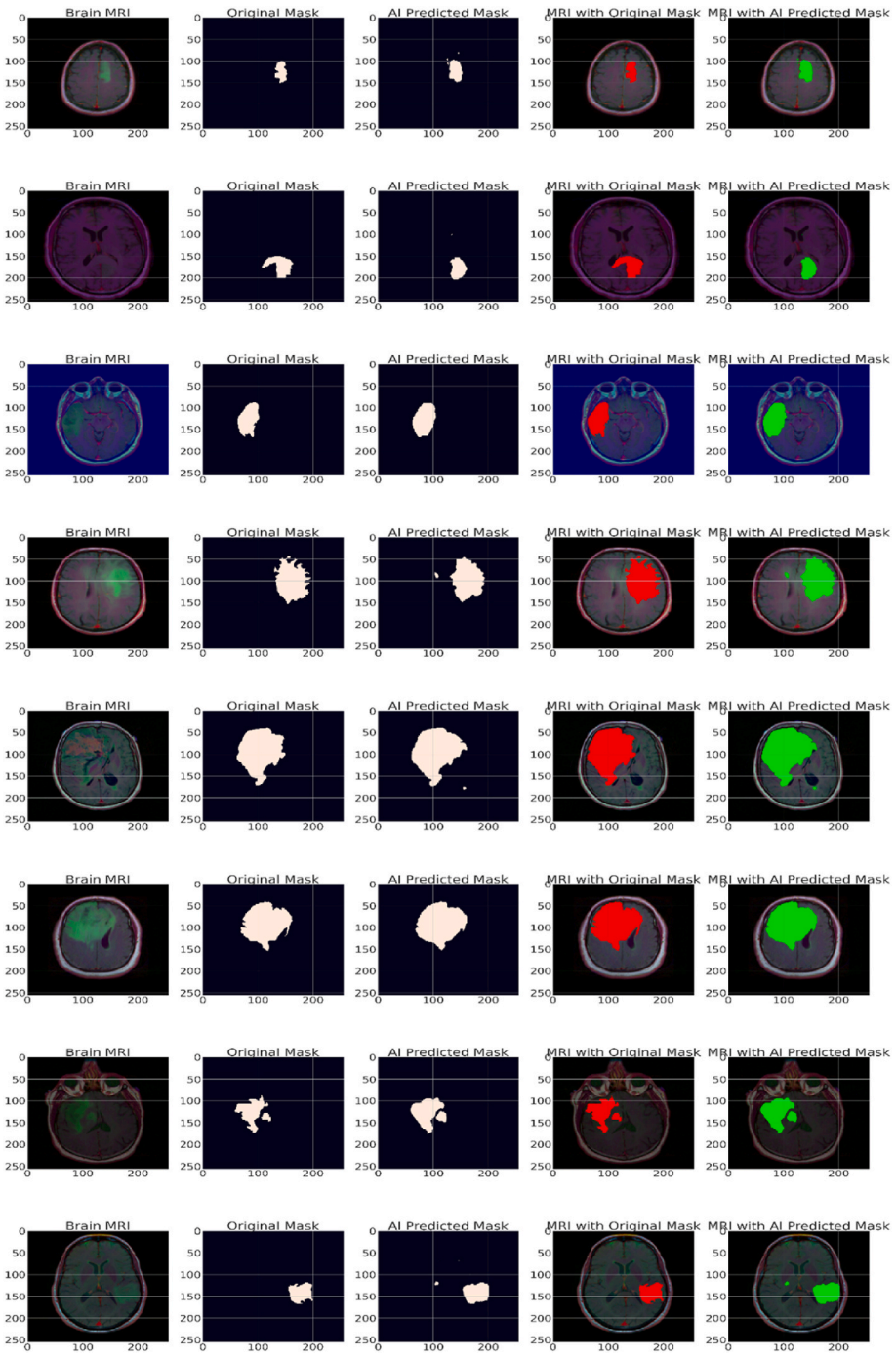


Fig. 23. Localization of tumor using MobileNet combined with Transfer Learning Layer and RESUNET prediction model.

overdiagnosis carries serious consequences. ResNet101 + Transfer Learning Layer, on the other hand, achieves the highest true positive rate and the lowest number of false negatives, making it the most sensitive model for tumor detection. Its downside is a higher number of false positives, which could lead to unnecessary tests or interventions. Therefore, ResNet101 + Transfer Learning Layer is preferable in scenarios where missing a tumor is unacceptable, even if it increases the risk of over-testing.

The proposed structure shows some interesting benefits that increase its efficiency in the location and categorisation of brain tumors. First it uses pre-trained models such MobileNet, NASNetMobile, and ResNet101 to enable efficient feature learning via transfer learning. These models especially help with generalisation, reduce training durations, and increase feature extraction capacity in circumstances with insufficient medical data. Second, the system provides consistent tumour localization by employing RESUNET—which combines residual connections with the U-Net design. This design allows precise boundary recognition and pixel-by- pixel segmentation of tumors patches, hence enhancing spatial precision.

Another advantage is the minimal weight and easy deployment of the structure. Thanks to architectures like MobileNet and NASNetMobile, which are highly optimized for speed and performance, the idea is suitable for real-time clinical applications and utilisation in resource-constrained conditions including embedded devices. Furthermore advantageous for the method is enhanced learning using multimodal MRI data. By combining many imaging modalities, including T1, T2, and

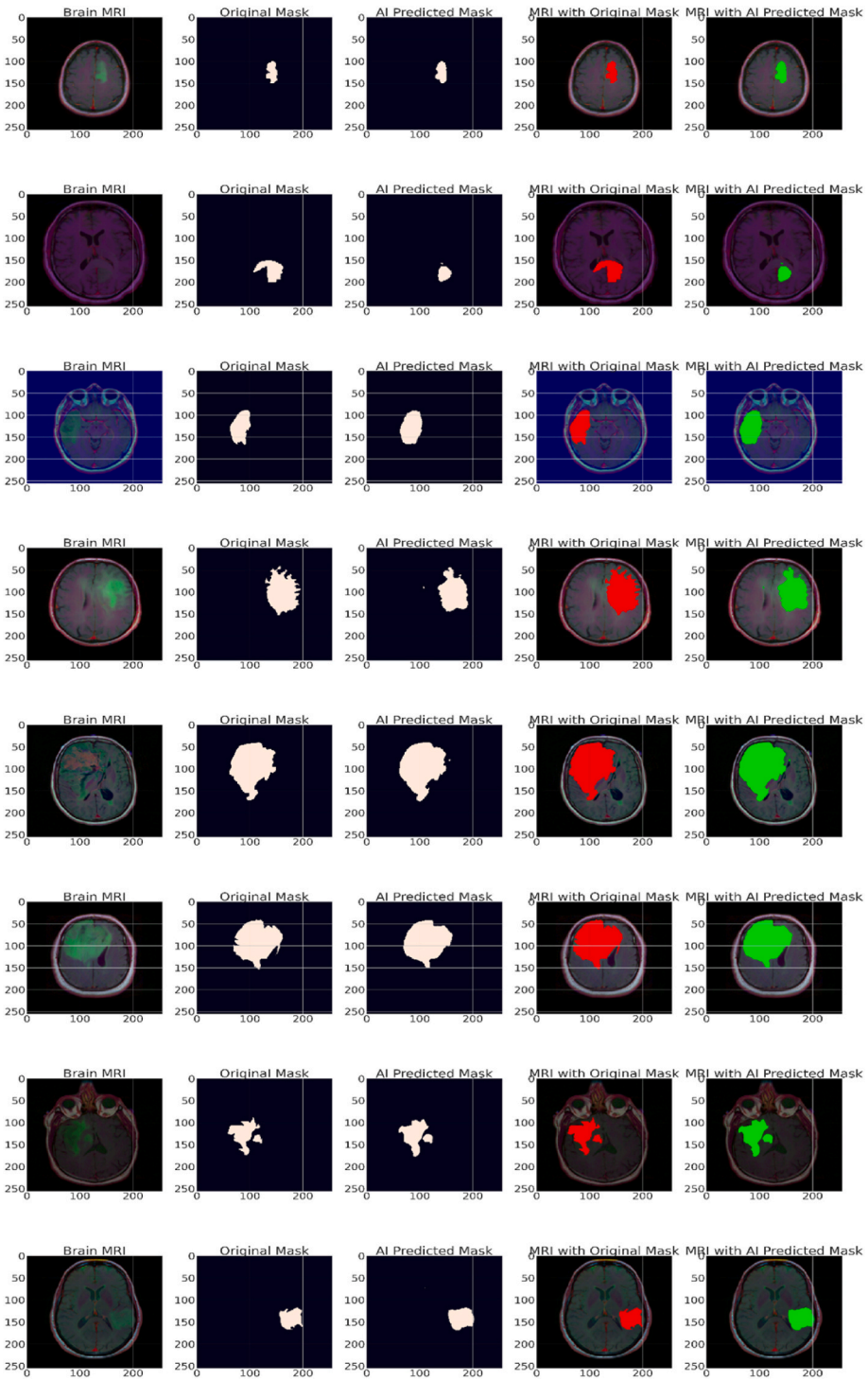


Fig. 24. Localization of tumor using NASNetMobile combined with Transfer Learning Layer and RESUNET prediction model.

FLAIR, the model gains a more complete awareness of tumour characteristics and produces more exact predictions.

A significant architectural strength is its unusual dual-stage design, which manages localization (tumour boundary mapping) and classification (tumour type detection) separately. This improves general performance and interpretability by allowing every step to be specifically suited for its intended use. Finally, the framework shows tremendous accuracy and outstanding generalisation abilities. Although NASNetMobile offers a reasonable compromise between model complexity and performance, ResNet101 enables deep feature learning—which is very vital in complex tumors. This lets the system be customised to meet many clinical needs.

The dual-stage deep learning system has limitations even if it offers notable improvements in the classification and localization of brain tumors. First, a significant obstacle still exists in the need for large, labeled, and multimodal MRI datasets. Because medical picture annotation is time-consuming and requires subject knowledge, there are few and expensive high-quality labeled datasets available. This has an impact on the model's capacity to generalise to previously unknown data, particularly when it is used at several institutions with disparate imaging techniques.

Accuracy may also be impacted by the domain shift between medical

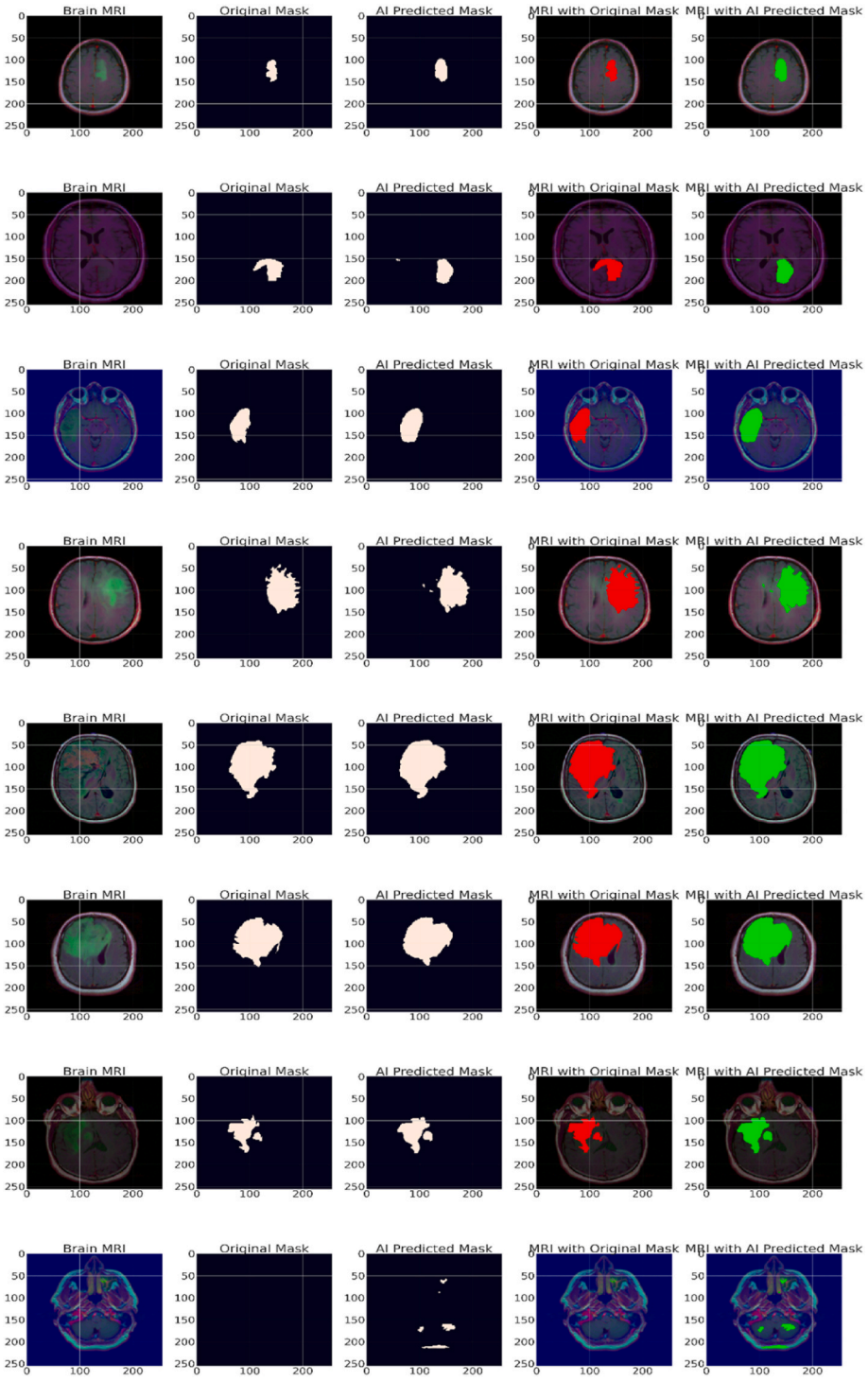


Fig. 25. Localization of tumor using ResNet101 combined with Transfer Learning Layer and RESUNET prediction model.

imaging data and pre-trained models. The distinct texture, contrast, and spatial patterns seen in MRI scans are not captured by pre-trained models like MobileNet or ResNet, which are trained on natural image datasets like ImageNet. Despite its benefits, transfer learning could still result in less-than-ideal performance unless it is followed by thorough domain-specific fine-tuning. The processing load and inference time are additional drawbacks, especially when merging different modalities or utilizing deeper models like ResNet101. This may restrict the framework's use in clinical settings with constrained computing resources and impede real-time analysis. Data alignment problems are also brought about by multimodal input integration, necessitating time-consuming and error-prone sophisticated preparation procedures including registration, normalization, and modality fusion.

In the context of MRI-based tumor detection using MobileNet + Transfer Learning Layer, NASNetMobile + Transfer Learning Layer, and ResNet101 + Transfer Learning Layer, misclassifications—either false positives (FP) or false negatives (FN)—can arise from several technical and clinical factors:

- MRI Intensity Variations or Noise: Even after preprocessing, MRIs may exhibit variations in intensity, artifacts, or random noise that can obscure subtle features of tumors. CNNs rely on consistent

Table 6
Comparison with other state-of-the-art methods.

S. No	Reference	Model and Framework	Accuracy
1	[44]	CNN-GAN	0.8653
2	[45]	CNN-SVM	0.9310
3	[45]	CNN-KNN	0.8750
4	[46]	UNet with RESNeXt50	0.9320
5	[47]	DenseNet-121	0.9210
6	[47]	GoogLeNet	0.8730
7	[47]	MobileNet	0.8890
8	[48]	QDCNN-DMN	0.9400
9	[49]	MSMCNN-LSTM	0.9636
10	[50]	Deep Convolutional Neural Networks	0.9650
11	[51]	UNet - DCNN	0.8860
12	[52]	ResNet-50	0.9530
13	Proposed	MobileNet + Transfer Learning Layer	0.9600
14	Proposed	NASNetMobile + Transfer Learning Layer	0.9700
15	Proposed	ResNet101 + Transfer Learning Layer	0.9500

Table 7
For each confusion matrix.

Model	True Negatives (TN)	False Positives (FP)	False Negatives (FN)	True Positives (TP)
MobileNet + Transfer Learning Layer	491	6	24	263
NASNetMobile + Transfer Learning Layer	493	4	20	267
ResNet101 + Transfer Learning Layer	478	19	17	270

patterns to classify images accurately, so these variations may lead the network to misinterpret normal tissue as tumor (FP) or miss actual tumor regions (FN).

- Small or Diffuse Tumors: Tumors that are very small, diffuse, or lack a well-defined boundary are inherently more difficult for convolutional neural networks to detect. These tumors may not generate strong activation in the feature maps of the CNN, increasing the likelihood of false negatives. Such cases are particularly challenging because they can be clinically significant, even if numerically rare.
- Limitations in Feature Extraction by the Backbone CNN: While MobileNet, NASNetMobile, and ResNet101 are powerful feature extractors, they may have limitations in capturing all the subtle spatial or textural features present in MRI data. For example, MobileNet's lightweight architecture may miss fine-grained details, whereas NASNetMobile might focus on macro patterns, and ResNet101, despite being deeper, may overemphasize certain patterns at the cost of others. These limitations can contribute to both FP and FN.
- Overfitting to Training Data: If the CNN models were trained on a dataset that is not fully representative of the variety of tumor presentations, the models may overfit to certain features. Overfitting could manifest as confident but incorrect predictions: labeling non-tumor tissue as tumor (FP) or failing to detect atypical tumors (FN). This is particularly relevant if tumor classes are heterogeneous, e.g., varying in size, location, or intensity.

Additionally, the framework has issues with clinical trust and interpretability. Hence some limitations of the Methodology are following.

- Limited Dataset Size and Diversity: With fewer than 4000 images, the dataset may not sufficiently capture the full variability of GBM presentation across patients, MRI scanners, and imaging protocols, potentially affecting model generalizability.
- Computational Complexity: The dual-stage framework—combining three CNN architectures with transfer learning and RESUNET segmentation—requires substantial computational resources, which may hinder deployment in hospitals or clinics without high-end GPUs.
- Risk of Overfitting: Deeper networks like ResNet101, even with transfer learning, are prone to overfitting, particularly given the relatively small and imbalanced dataset.
- Limited Generalizability: The model was trained and validated solely on TCGA-GBM images. Performance on other datasets, multi-institutional data, or real-world clinical images is untested, raising concerns about external validity.
- Integration Complexity: Combining classification and segmentation increases system complexity, making implementation, maintenance, and reproducibility more challenging for clinical practitioners.
- Dependence on Pretrained Weights: Transfer learning relies on pre-trained weights from non-medical datasets, which may not capture domain-specific MRI features optimally, potentially limiting feature extraction effectiveness.
- Scalability Challenges: Extending the framework to multi-class or multi-modality MRI classification would increase computational demands and may require re-architecting the dual-stage pipeline.
- Even with RESUNET's visible outputs, deep models' core decision-making is still unclear. This lack of transparency may erode physician trust and impede adoption in high-stakes settings like healthcare. This might be lessened by integrating explainable AI techniques, albeit these methods are still developing.

7. Conclusion

This study presents a dual-stage deep learning framework for automated brain tumor analysis using multimodal MRI images from the TCGA-GBM dataset. The system integrates a classification stage with a localization stage, combining Transfer Learning-based models—MobileNet, NASNetMobile, and ResNet101—with a RESUNET segmentation architecture. In the classification stage, NASNetMobile + Transfer Learning Layer achieved the highest balance of accuracy (0.9700) and computational efficiency, MobileNet + Transfer Learning Layer provided lightweight yet competitive performance (accuracy 0.9600), and ResNet101 + Transfer Learning Layer delivered the highest sensitivity for tumor detection (accuracy 0.9500) at the cost of higher computational demands. In the localization stage, coupling these models with RESUNET enabled precise tumor delineation, highlighting the potential to assist radiologists in both detection and planning for intervention. This dual-stage framework demonstrates that high accuracy and efficiency can be achieved simultaneously, offering a scalable solution for automated MRI analysis in clinical workflows. The practical significance of this work lies in its ability to reduce diagnostic time, standardize tumor detection, and support decision-making in healthcare settings. Furthermore, the framework is adaptable, allowing future enhancements such as inclusion of additional MRI modalities, incorporation of advanced architectures like transformers or attention mechanisms, and validation on multi-institutional datasets to improve generalizability. Long-term goals include real-time deployment and collaborative integration with medical professionals, ensuring safe, efficient, and clinically meaningful application of AI-driven tumor analysis.

CRediT authorship contribution statement

Deependra Rastogi: Methodology. **Prashant Johri:** Methodology. **Sumit Singh Dhanda:** Conceptualization. **Anand Singh:** Conceptualization. **Suman Avdhesh Yadav:** Formal analysis. **Arfat Ahmad Khan:** Resources. **Seifedine Kadry:** Investigation.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Seifedine Kadry reports was provided by Lebanese American University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset used in this study, " Pedano, N., Flanders, A. E., Scarpace, L., Mikkelsen, T., Eschbacher, J. M., Hermes, B., Sisneros, V., Barnholtz-Sloan, J., & Ostrom, Q. (2016). The Cancer Genome Atlas Low Grade Glioma Collection (TCGA-LGG) (Version 3) [Data set]. The Cancer Imaging Archive. https://doi.org/10.7937/K9/TCIA.2016.L4LTD3TK"

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